



DATABASE FOR HOSPITAL

IT 131 - 3300

Dr.Abeer Alhujaylan

Ghadah Mujahid - Lubna Albusayr - Bana AlTuwaijri

Renad AlMohsen - Dina AlAjlan - Aseel AlEid - Ruba AlAjlan - kadi AlSubaie

Entities

◆ *Patient*

◆ *Medical Record*

◆ *Clinical Laboratory*

◆ *Doctor*

◆ *Appointments*

◆ *Patient Feedback*

◆ *Nurse*

◆ *Department*

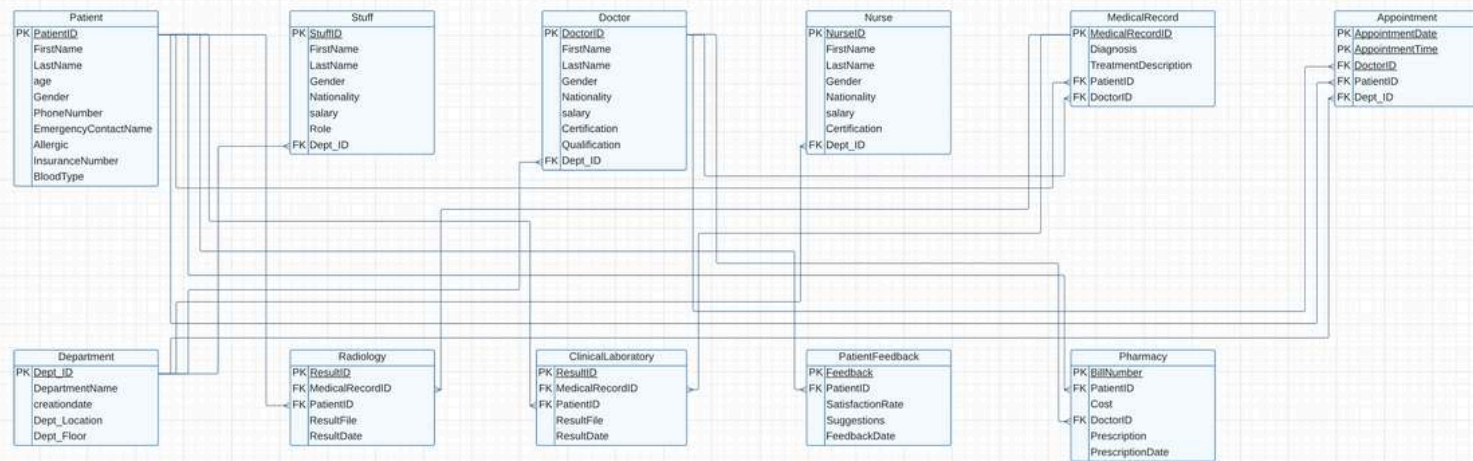
◆ *Pharmacy*

◆ *Staff*

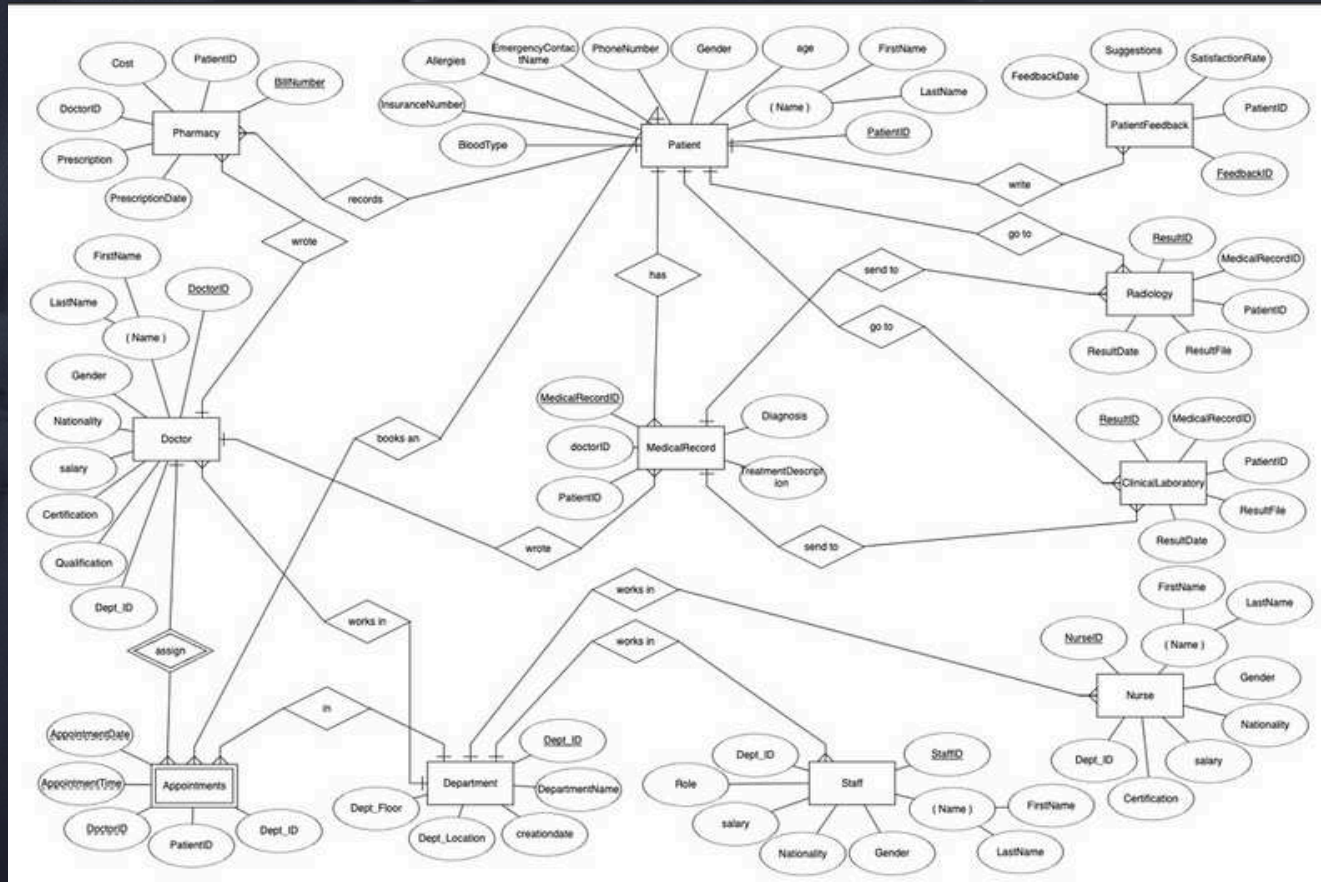
◆ *Radiology*



The Schema



The ER-Diagram



DDL (Creation)

```
CREATE TABLE Patient (  
  PatientID INT PRIMARY KEY,  
  FirstName VARCHAR(50) NOT NULL,  
  LastName VARCHAR(50) NOT NULL,  
  age INT,  
  Gender CHAR(1) NOT NULL,  
  PhoneNumber VARCHAR(15) not null,  
  EmergencyContactName VARCHAR(100),  
  Allergies VARCHAR(255),  
  InsuranceNumber VARCHAR(50) ,  
  BloodType CHAR(3) NOT NULL  
);
```

Create table
Patient

Create table
Doctor

```
CREATE TABLE Doctor (  
  DoctorID INT PRIMARY KEY,  
  FirstName VARCHAR(50) NOT NULL,  
  LastName VARCHAR(50) NOT NULL,  
  Gender CHAR(1) NOT NULL,  
  Nationality VARCHAR(50) NOT NULL,  
  salary DECIMAL(10, 2) NOT NULL,  
  Certification VARCHAR(100) NOT NULL,  
  Qualification VARCHAR(100) NOT NULL,  
  Dept_ID INT NOT NULL,  
  FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID)  
);
```

```
CREATE TABLE Nurse (  
  NurseID INT PRIMARY KEY,  
  FirstName VARCHAR(50) NOT NULL,  
  LastName VARCHAR(50) NOT NULL,  
  Gender CHAR(1) NOT NULL,  
  Nationality VARCHAR(50) NOT NULL,  
  salary DECIMAL(10, 2) NOT NULL,  
  Certification VARCHAR(100) NOT NULL,  
  Dept_ID INT NOT NULL,  
  FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID)  
);
```

Create table Nurse

Create table Staff

```
CREATE TABLE Staff (  
  StaffID INT PRIMARY KEY,  
  FirstName VARCHAR(50) NOT NULL,  
  LastName VARCHAR(50) NOT NULL,  
  Gender CHAR(1) NOT NULL,  
  Nationality VARCHAR(50) NOT NULL,  
  salary DECIMAL(10, 2) NOT NULL,  
  Role VARCHAR(50) NOT NULL,  
  Dept_ID INT NOT NULL,  
  FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID)  
);
```

```
CREATE TABLE MedicalRecord (  
  MedicalRecordID INT PRIMARY KEY,  
  Diagnosis VARCHAR(255) NOT NULL,  
  TreatmentDescription VARCHAR(255) NOT NULL,  
  PatientID INT NOT NULL,  
  doctorID INT NOT NULL,  
  FOREIGN KEY (PatientID) REFERENCES Patient(PatientID),  
  FOREIGN KEY (doctorID) REFERENCES Doctor(DoctorID)  
);
```

Create table MedicalRecord

```
CREATE TABLE Appointments (  
  AppointmentDate DATE NOT NULL,  
  AppointmentTime TIME NOT NULL,  
  DoctorID INT NOT NULL,  
  PatientID INT NOT NULL,  
  Dept_ID INT NOT NULL,  
  PRIMARY KEY (AppointmentDate, AppointmentTime, DoctorID),  
  FOREIGN KEY (PatientID) REFERENCES Patient(PatientID),  
  FOREIGN KEY (DoctorID) REFERENCES Doctor(DoctorID),  
  FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID)  
);
```

Create table
Appointments

Create table
Department

```
CREATE TABLE Department (  
  Dept_ID INT PRIMARY KEY,  
  DepartmentName VARCHAR(50) NOT NULL,  
  creationdate date not null,  
  Dept_Location VARCHAR(50),  
  Dept_Floor INT,  
);
```

Create table Clinical Laboratory

```
CREATE TABLE ClinicalLaboratory (  
  ResultID INT PRIMARY KEY,  
  MedicalRecordID INT NOT NULL,  
  PatientID INT NOT NULL,  
  ResultFile VARCHAR(255) NOT NULL,  
  ResultDate DATE NOT NULL,  
  FOREIGN KEY (MedicalRecordID) REFERENCES MedicalRecord(MedicalRecordID),  
  FOREIGN KEY (PatientID) REFERENCES Patient(PatientID)  
);
```

Create table Radiology

```
CREATE TABLE Radiology (  
  ResultID INT PRIMARY KEY,  
  MedicalRecordID INT NOT NULL,  
  PatientID INT NOT NULL,  
  ResultFile VARCHAR(255) NOT NULL,  
  ResultDate DATE NOT NULL,  
  FOREIGN KEY (MedicalRecordID) REFERENCES MedicalRecord(MedicalRecordID),  
  FOREIGN KEY (PatientID) REFERENCES Patient(PatientID)  
);
```

```
CREATE TABLE PatientFeedback (  
  FeedbackID INT PRIMARY KEY,  
  PatientID INT NOT NULL,  
  SatisfactionRate INT NOT NULL,  
  Suggestions VARCHAR(255),  
  FeedbackDate DATE NOT NULL,  
  FOREIGN KEY (PatientID) REFERENCES Patient(PatientID)  
);
```

Create table
PatientFeedback

Create table
Pharmacy

```
CREATE TABLE Pharmacy (  
  BillNumber INT PRIMARY KEY,  
  PatientID INT NOT NULL,  
  Cost DECIMAL(10, 2),  
  DoctorID INT NOT NULL,  
  Prescription VARCHAR(255) NOT NULL,  
  PrescriptionDate DATE NOT NULL,  
  FOREIGN KEY (PatientID) REFERENCES Patient(PatientID),  
  FOREIGN KEY (DoctorID) REFERENCES Doctor(DoctorID)  
);
```

DML (insertion) Patient

```
insert into Patient
values (110101,'Lana','Ahmead',28,'F','0505587433','Ahmead Nasser','Peanut','111-021-001','0+'),
(110102,'Lamar','Fahad',18,'F','0504599344','Waleed Fahad','No Allergies','112-022-221','0+'),
(110103,'Naif','Saleh',12,'M','0505170444','Saleh Ali','Soya','111-221,920','A+'),
(110104,'Leen','Sultan',17,'F','0405735728','Kadi Ali','Fish','No Insurance','0+'),
(110105,'Sultan','Mohammed',45,'M','0405735728','Asma Khalid','No Allergies','No Insurance','B+'),
(110106,'Dana','Mohamed',13,'F','0505464952','Nehad Abdullah','Eggs','112-989-030','AB+'),
(110107,'Hussam','Ali',34,'M','0500630222','Btool Ali','Fish','115-595-001','A+'),
(110108,'Arwa','Hamed',22,'F','0506146434','Nora Azzam','No Allergies','No Insurance','B-'),
(110109,'Rand','Fasial',5,'F','0506646433','Fasial Saud','Peanut','221-221-921','0+'),
(110110,'Bayan','Salman',32,'F','0505446443','Nawaf Fasial','No Allergies','224-443-343','0+'),
(110111,'Firas','Fasial',0,'M','0505446443','Bayan Salman','No Allergies','224-443-343','0+'),
(110112,'Abdullah','Rsheed',25,'M','0568850625','Rsheed Talal','Melon','No Insurance','B+'),
(110113,'Saleh','Ahemad',65,'M','0504894258','Noor Saleh','No Allergies','992-232-040','A+'),
(110114,'Luluah','Ibrahim',85,'F','0539440445','Bedoor Saleh','No Allergies','119-083-321','AB+'),
(110115,'Nasser','Saleh',44,'M','0503023203','Nada Fahad','Shellfish','119-083-321','AB+'),
(110116,'Leena','Ahemad',8,'F','0538815577','Ahemad Khalid','No Allergies','No Insurance','B+'),
(110117,'Yousef','Zaid',52,'M','0501804638','Maha Naif','No Milk','662-910-123','0+'),
(110118,'Manar','Rayan',20,'F','0556794717','Rayan Abdulaziz','Milk','No Insurance','A-'),
(110119,'Omar','Osamah',15,'M','0537665745','Osamah Turki','No Allergies','506-911-128','B+'),
(110120,'Marah','Saad',17,'F','0505834258','Mona Jamal','Fish','No Insurance','A+'),
(110121,'Haya','Rasheed',45,'F','050743296834','Talal Turki','No Allergies','No Insurance','A+'),
(110122,'Nawaf','Rakan',12,'M','0501693371','Rakan Naif','Fish','222-934-848','B+'),
(110123,'Luna','Sultan',5,'F','0505587433','Lana Ahemed','No Allergies','111-021-001','B+'),
(110124,'Yahya','Bader',20,'M','0505624354','Bader Zaid','Sesame Seeds','757-983-003','AB+'),
(110125,'Sattam','Salman',34,'M','0505395532','Muneraah Salman','Sesame Seeds','No Insurance','A-'),
(110126,'Huda','Tamim',18,'F','0508974693','Tamim Nasser','Bananas','No Insurance','B+'),
(110127,'Osamah','Husham',35,'M','0505983665','Drra Husham','Eggs','No Insurance','A+'),
(110128,'Fahad','Abdulaziz',18,'M','0505564422','Abdulaziz Saleh','Kiwis','637-098-092','AB+'),
(110129,'Asmaa','Omar',49,'F','0553893275','Kadi Ibrahim','No Allergies','983-993-001','AB+'),
(110130,'Hussam','Ibrahim',10,'M','0553893275','Asmaa Omar','Kiwis','983-993-001','B+'),
(110131,'Yousef','Majed',49,'M','0509712764','Majed Yousef','No Allergies','No Insurance','A+'),
(110132,'Maha','Ibrahim',25,'F','0557632211','Ibrahim Nawaf','Peanut','No Insurance','0+'),
(110133,'Khalid','Abdelrahman',29,'M','0505067433','Abdelrahman Saud','Peanut','999-878-444','0+'),
(110134,'Saud','Fars',75,'M','0505058988','Farah Fars','No Allergies','No Insurance','B+'),
(110135,'Farah','Isa',18,'F','0505999887','Isa Mohammed','Peanut','999-000-123','0+'),
(110136,'Marwan','Malk',32,'M','0509871234','Malk Mohammed','No Allergies','No Insurance','B+'),
(110137,'Lama','Abdullah',17,'F','0508932278','Abdullah Saud','Strawberries','876-999-003','AB+'),
(110138,'Hussa','Tariq',53,'F','0508974666','Dania Fars','No Allergies','No Insurance','0+'),
(110139,'Suleiman','Hamed',27,'M','0500057489','Hamed Rakan','Strawberries','No Insurance','B-'),
(110140,'Saad','Malk',16,'M','0509876432','Malk Khalid','Peanut','No Insurance','A+');
```

Patient

	PatientID	FirstName	LastName	age	Gender	PhoneNumber	EmergencyContactName	Allergies	InsuranceNumber	BloodType
1	110101	Lana	Ahmed	28	F	0505587433	Ahmed Nasser	Peanut	111-021-001	O+
2	110102	Lamar	Fahad	18	F	0504599344	Waleed Fahad	No Allergies	112-022-221	O+
3	110103	Naif	Saleh	12	M	0505170444	Saleh Ali	Soya	111-221,920	A+
4	110104	Leen	Sultan	17	F	0405735728	Kadi Ali	Fish	No Insurance	O+
5	110105	Mohamdan	Mohammed	45	M	0405735728	Asma Khalid	No Allergies	No Insurance	B+
6	110106	Dana	Mohamed	13	F	0505464952	Nehad Abdullah	Eggs	112-989-030	AB+
7	110107	Hussam	Ali	34	M	0500630222	Btool Ali	Fish	115-595-001	A+
8	110108	Arwa	Hamed	22	F	0506146434	Nora Azzam	No Allergies	No Insurance	B-
9	110109	Rand	Fasial	5	F	0506646433	Fasial Saud	Peanut	221-221-921	O+
10	110110	Bayan	Salman	32	F	0505446443	Nawaf Fasial	No Allergies	224-443-343	O+
11	110111	Firas	Fasial	0	M	0505446443	Bayan Salman	No Allergies	224-443-343	O+
12	110112	Abdullah	Rsheed	25	M	0568850625	Rsheed Talal	Melon	No Insurance	B+
13	110113	Saleh	Ahmed	65	M	0504894258	Noor Saleh	No Allergies	992-232-040	A+
14	110114	Luluah	Ibrahim	85	F	0539440445	Bedoor Saleh	No Allergies	119-083-321	AB+
15	110115	Nasser	Saleh	44	M	0503023203	Nada Fahad	Shellfish	119-083-321	AB+
16	110116	Leena	Ahmed	8	F	0538815577	Ahmed Khalid	No Allergies	No Insurance	B+
17	110117	Yousef	Zaid	52	M	0501804638	Maha Naif	No Milk	662-910-123	O+
18	110118	Manar	Rayan	20	F	0556794717	Rayan Abdulaziz	Milk	No Insurance	A-
19	110119	Omar	Osamah	15	M	0537665745	Osamah Turki	No Allergies	506-911-128	B+
20	110120	Marah	Saad	17	F	0505834258	Mona Jamal	Fish	No Insurance	A+
21	110121	Haya	Rasheed	45	F	050743296834	Talal Turki	No Allergies	No Insurance	A+
22	110122	Nawaf	Rakan	12	M	0501693371	Rakan Naif	Fish	222-934-048	B+
23	110123	Luna	Sultan	5	F	0505587433	Lana Ahmed	No Allergies	111-021-001	B+
24	110124	Yahya	Bader	20	M	0505624354	Bader Zaid	Sesame Seeds	757-983-003	AB+
25	110125	Sattam	Salman	34	M	0505395532	Muneraah Salman	Sesame Seeds	No Insurance	A-
26	110126	Huda	Tamim	18	F	0508974693	Tamim Nasser	Bananas	No Insurance	B+
27	110127	Osamah	Husham	35	M	0505983665	Drira Husham	Eggs	No Insurance	A+
28	110128	Fahad	Abdulaziz	18	M	0505564422	Abdulaziz Saleh	Kiwis	637-090-092	AB+
29	110129	Asmaa	Omar	49	F	0553093275	Kadi Ibrahim	No Allergies	903-993-001	AB+
30	110130	Hussam	Ibrahim	10	M	0553093275	Asmaa Omar	Kiwis	903-993-001	B+
31	110131	Yousef	Majed	49	M	0509712764	Majed Yousef	No Allergies	No Insurance	A+
32	110132	Maha	Ibrahim	25	F	0557632211	Ibrahim Nawaf	Peanut	No Insurance	O+
33	110133	Khalid	Abdelrahman	29	M	0505067433	Abdelrahman Saud	Peanut	999-878-444	O+
34	110134	Saud	Fars	75	M	0505058988	Farah Fars	No Allergies	No Insurance	B+
35	110135	Farah	Isa	18	F	0505999887	Isa Mohammed	Peanut	999-000-123	O+
36	110136	Marwan	Malk	32	M	0509871234	Malk Mohammed	No Allergies	No Insurance	B+
37	110137	Lama	Abdullah	17	F	0508932278	Abdullah Saud	Strawberries	876-999-003	AB+
38	110138	Hussa	Tariq	53	F	0508974666	Dania Fars	No Allergies	No Insurance	O+
39	110139	Suleiman	Hamed	27	M	0500057489	Hamed Rakan	Strawberries	No Insurance	B-
40	110140	Saad	Malk	16	M	0509876432	Malk Khalid	Peanut	No Insurance	A+

Doctor

INSERT INTO Doctor

```
VALUES (2201,'Diana','Geisser','F','American',40000,'MD,ABIM','Cardiologist',00058),
(2202,'Majed','Yaseer','M','Saudi',38000,'MD','Paediatric Cardiologist',00058),
(2203,'Tala','Ahemad','F','Saudi',30000,'MD','Cardiologist',00058),
(2204,'George','Chalkiadis','M','Saudi',60000,'MD,BLS','Anesthesiologist',03208),
(2205,'Saud','Turki','M','Saudi',56000,'MD,ABA','Consultant Anaesthetist',03208),
(2206,'Marwa','Osamah','F','Saudi',35000,'MD','Haematologist',02055),
(2207,'Layan','Sami','F','Syrian',40000,'MD','Consultant Haematologist',02055),
(2208,'Marah','Abdulkarim','F','Saudi',60000,'MD,ABD','Dermatologists',03137),
(2209,'Paul','Curnow','M','American',50000,'MD','Dermatologists',03137),
(2210,'Kathleen','McGrath','F','American',28000,'DO','Gastroenterology',01048),
(2211,'Naif','Faisal','M','Saudi',32000,'MD,ABIM','Gastroenterology',01048),
(2212,'Sarah','Curnow','F','British',34000,'MD','Neurologists',06044),
(2213,'Michael','Hayman','F','British',34000,'MD','Neurologists',06044),
(2214,'Rania','Rasheed','F','Saudi',25000,'CNS,RDN','Nutrition Consultant',00203),
(2215,'Rayoof','Khalid','F','Saudi',25000,'CNS','Nutrition Consultant',00203),
(2216,'Saleh','Naif','M','Saudi',28000,'MBBS, FRANZCO','Ophthalmologist',00046),
(2217,'Wendy','Marshman','F','American',32000,'MBBS, MD, FRANZCO, FRACS','Ophthalmologist',00046),
(2218,'Baseel','Tariq','M','Saudi',29000,'MBBS, FRANZCO','Ophthalmologist',00046),
(2219,'Mariam','Shaker','F','Pakistani',31000,'ENT,MD','Otolaryngologist',02042),
(2220,'Sam','Davies','M','American',36000,'ENT,MD','Otolaryngologist',02042),
(2221,'Rada','Saleh','F','Saudi',30000,'DPT','Doctor Of Physical Therapy',00119),
(2222,'Maha','Ali','F','Saudi',30000,'DPT','Doctor Of Physical Therapy',00119),
(2223,'Adeel','Rakan','M','Saudi',34000,'Bachelors degree in Psychology,MSP','Therpaist',01202),
(2224,'Zaid','Marwan','M','Saudi',34000,'Bachelors degree in Psychology,MAP','Therpaist',01202),
(2225,'John','Hutson','M','British',45000,'MD','Surgeon',00058),
(2226,'Juan','Bortagaray','M','British',45000,'MD','Surgeon',00058),
(2227,'Taraf','Abdelmohsen','F','Saudi',50000,'MD,FRACS','Surgeon',03137),
(2228,'Dalal','Abdalmajeed','F','Saudi',48000,'MD,FAAP','Surgeon',03137),
(2229,'David','Francis','M','American',53000,'BS,MD,DSc,FRACS, FAAP','Surgeon',04057),
(2230,'Shahad','Waleed','F','Egyptian',43000,'MD,ABU','Urologist',04057),
(2231,'Saber','Ramadan','M','Pakistani',33000,'MD','Urologist',04057),
(2232,'Azurah','Sami','F','Egyptian',45000,'MD,GYN,REI','Gynaecologist',00130),
(2233,'Mariam','Hussen','F','Saudi',44000,'MD,MIGS','Gynaecologist',00130),
(2234,'Kayne','Cornelis','M','American',40000,'DDS,MSc,PHD','Orthodontists',03111),
(2235,'Mashal','Abdulkarim','M','Saudi',40000,'BDSc, MDSc, MRACDS (Ortho), FRACDS','Orthodontists',03111),
(2236,'Jumanah','Mohammed','F','Saudi',44000,'BDSc, MRACDS (GDP)','Dental',03111),
(2237,'Ashraf','Hamad','M','Saudi',30000,'BHSc, MDent','Dental',03111),
(2238,'Mohammed','Abdelrahman','M','Saudi',54000,'MD','Nephrologist',06044),
(2239,'Ling','Chen','F','Chinese',54000,'MD','Nephrologist',06044),
(2240,'Rashed','Zyad','M','Saudi',51000,'MD','Oncologist',03045),
(2241,'Reem','Yousef','F','Saudi',50000,'MD','Oncologist',03045);
```

Doctor

	DoctorID	FirstName	LastName	Gender	Nationality	salary	Certification	Qualification	Dept_ID
1	2201	Diana	Geisser	F	American	40000.00	MD, ABIM	Cardiologist	58
2	2202	Majed	Yaseer	M	Saudi	30000.00	MD	Paediatric Cardiologist	58
3	2203	Tala	Ahenad	F	Saudi	30000.00	MD	Cardiologist	58
4	2204	George	Chalkiadis	M	Saudi	60000.00	MD, BLS	Anesthesiologist	3208
5	2205	Saud	Turki	M	Saudi	56000.00	MD, ABA	Consultant Anaesthetist	3208
6	2206	Marwa	Osenah	F	Saudi	35000.00	MD	Haematologist	2055
7	2207	Layan	Sami	F	Syrian	40000.00	MD	Consultant Haematologist	2055
8	2208	Marah	Abdulkarin	F	Saudi	60000.00	MD, ABD	Dermatologists	3137
9	2209	Paul	Curnow	M	American	50000.00	MD	Dermatologists	3137
10	2210	Kathleen	McGrath	F	American	20000.00	DO	Gastroenterology	1048
11	2211	Naif	Faisal	M	Saudi	32000.00	MD, ABIM	Gastroenterology	1048
12	2212	Sarah	Curnow	F	British	34000.00	MD	Neurologists	6044
13	2213	Michael	Hayman	F	British	34000.00	MD	Neurologists	6044
14	2214	Rania	Rosheed	F	Saudi	25000.00	CNS, RDW	Nutrition Consultant	203
15	2215	Rayoof	Khalid	F	Saudi	25000.00	CNS	Nutrition Consultant	203
16	2216	Saleh	Naif	M	Saudi	28000.00	MBBS, FRANZCO	Ophthalmologist	46
17	2217	Wendy	Marshman	F	American	32000.00	MBBS, MD, FRANZCO, FRACS	Ophthalmologist	46
18	2218	Baseel	Tariq	M	Saudi	29000.00	MBBS, FRANZCO	Ophthalmologist	46
19	2219	Marian	Shaker	F	Pakistani	31000.00	ENT, MD	Otolaryngologist	2042
20	2220	Sam	Davies	M	American	36000.00	ENT, MD	Otolaryngologist	2042
21	2221	Rada	Saleh	F	Saudi	30000.00	DPT	Doctor Of Physical Therapy	119
22	2222	Maha	Ali	F	Saudi	30000.00	DPT	Doctor Of Physical Therapy	119
23	2223	Adeel	Rakan	M	Saudi	34000.00	Bachelors degree in Psychology, MSP	Therpaist	1202
24	2224	Zaid	Marwan	M	Saudi	34000.00	Bachelors degree in Psychology, MAP	Therpaist	1202
25	2225	John	Hutson	M	British	45000.00	MD	Surgeon	58
26	2226	Juan	Bortagaray	M	British	45000.00	MD	Surgeon	58
27	2227	Taraf	Abdelmohsen	F	Saudi	50000.00	MD, FRACS	Surgeon	3137
28	2228	Dalal	Abdalmajeed	F	Saudi	40000.00	MD, FAAP	Surgeon	3137
29	2229	David	Francis	M	American	53000.00	BS, MD, DSc, FRACS, FAAP	Surgeon	4057
30	2230	Shahad	Waleed	F	Egyptian	43000.00	MD, ABU	Urologist	4057
31	2231	Saber	Ramadan	M	Pakistani	33000.00	MD	Urologist	4057
32	2232	Azurah	Sami	F	Egyptian	45000.00	MD, GYN, REI	Gynaecologist	130
33	2233	Marian	Hussen	F	Saudi	44000.00	MD, MIGS	Gynaecologist	130
34	2234	Kayne	Cornelis	M	American	40000.00	DOS, MSc, PhD	Orthodontists	3111
35	2235	Mashal	Abdulkarin	M	Saudi	40000.00	BDS, MSc, MRACDS (Ortho), FRACDS	Orthodontists	3111
36	2236	Jumanah	Mohammed	F	Saudi	44000.00	BDS, MRACDS (GDP)	Dental	3111
37	2237	Ashraf	Hamad	M	Saudi	30000.00	BHSc, MDent	Dental	3111
38	2238	Mohammed	Abdelrahman	M	Saudi	54000.00	MD	Nephrologist	6044
39	2239	Ling	Chen	F	Chinese	54000.00	MD	Nephrologist	6044
40	2240	Rashed	Zyad	M	Saudi	51000.00	MD	Oncologist	3045
41	2241	Reem	Yousef	F	Saudi	50000.00	MD	Oncologist	3045

Nurse

```
INSERT INTO Nurse  
VALUES
```

```
(123201, 'Emily', 'Johnson', 'F', 'American', 20000, 'Certified Registered Nurse Anesthetist', 03208),  
(123202, 'Michael', 'Garcia', 'M', 'American', 22000, 'Certified Registered Nurse Anesthetist', 03208),  
(125801, 'Sophia', 'Jad', 'F', 'Egyptian', 22000, 'Certified Cardiac Nurse', 00058),  
(125802, 'Nada', 'Osamah', 'F', 'Saudi', 21000, 'Certified Cardiac Nurse', 00058),  
(125501, 'Loui', 'Abdelkader', 'M', 'Saudi', 25000, 'Certified Hematology Nurse', 02055),  
(125502, 'Husham', 'Fars', 'M', 'Saudi', 26000, 'Certified Hematology Nurse', 02055),  
(125701, 'Amal', 'Ameer', 'F', 'Saudi', 24000, 'Certified Urology Nurse', 04057),  
(125702, 'Hazm', 'Adeeb', 'M', 'Saudi', 22000, 'Certified Urology Nurse', 04057),  
(123001, 'Rama', 'Ali', 'F', 'Egyptian', 26000, 'Certified Gynecology Nurse', 00130),  
(123002, 'Joury', 'Hani', 'F', 'Saudi', 24000, 'Certified Gynecology Nurse', 00130),  
(123701, 'Ahad', 'Yousef', 'F', 'Saudi', 24000, 'Certified Dermatology Nurse', 03137),  
(123702, 'Maria', 'Mazn', 'F', 'Saudi', 21000, 'Certified Dermatology Nurse', 03137),  
(124601, 'Mazn', 'Hmood', 'M', 'Saudi', 23000, 'Certified Ophthalmic Nurse', 00046),  
(124602, 'Aram', 'Abdulaziz', 'F', 'Saudi', 22000, 'Certified Ophthalmic Nurse', 00046),  
(124801, 'Draa', 'Talal', 'F', 'Saudi', 22000, 'Certified Gastroenterology Nurse', 01048),  
(124802, 'Abdelkader', 'Aimen', 'M', 'Saudi', 25000, 'Certified Gastroenterology Nurse', 01048),  
(124201, 'Mashal', 'Hussen', 'M', 'Saudi', 23000, 'Certified Otolaryngology Nurse', 02042),  
(124202, 'Sarah', 'Saleh', 'F', 'Saudi', 23000, 'Certified Otolaryngology Nurse', 02042),  
(124501, 'Ghadi', 'Ahmed', 'F', 'Saudi', 20000, 'Certified Oncology Nurse', 03045),  
(124502, 'Albtool', 'Eyad', 'F', 'Saudi', 21000, 'Certified Oncology Nurse', 03045),  
(124401, 'Fouz', 'Ibrahim', 'F', 'Saudi', 22000, 'Certified Neurology Nurse', 06044),  
(124402, 'Selena', 'Smith', 'F', 'American', 20000, 'Certified Neurology Nurse', 06044),  
(120301, 'Emma', 'Gomez', 'F', 'American', 21000, 'Certified Nutrition and Dietetics Nurse', 00203),  
(120302, 'Emily', 'Johnson', 'F', 'American', 23000, 'Certified Nutrition and Dietetics Nurse', 00203),  
(120201, 'Sophia', 'Khan', 'F', 'Pakistani', 23000, 'Certified Psychiatric Nurse', 01202),  
(120202, 'Ariana', 'Smith', 'F', 'American', 21000, 'Certified Psychiatric Nurse', 01202),  
(121101, 'Maria', 'Chang', 'F', 'Chinese', 22000, 'Certified Dental Nurse', 03111),  
(121102, 'Michael', 'Nguyen', 'M', 'Vietnamese', 23000, 'Certified Dental Nurse', 03111),  
(121901, 'Hazza', 'Faisal', 'M', 'Saudi', 22000, 'Certified Physiotherapy Nurse', 00119),  
(121902, 'Husham', 'Hussen', 'M', 'Saudi', 22000, 'Certified Physiotherapy Nurse', 00119);
```

Nurse

	NurseID	FirstName	LastName	Gender	Nationality	salary	Certification	Dept_ID
1	120201	Sophia	Khan	F	Pakistani	23000.00	Certified Psychiatric Nurse	1202
2	120202	Ariana	Smith	F	American	21000.00	Certified Psychiatric Nurse	1202
3	120301	Emma	Gomez	F	American	21000.00	Certified Nutrition and Dietetics Nurse	203
4	120302	Emily	Johnson	F	American	23000.00	Certified Nutrition and Dietetics Nurse	203
5	121101	Maria	Chang	F	Chinese	22000.00	Certified Dental Nurse	3111
6	121102	Michael	Nguyen	M	Vietnamese	23000.00	Certified Dental Nurse	3111
7	121901	Hazza	Faisal	M	Saudi	22000.00	Certified Physiotherapy Nurse	119
8	121902	Husham	Hussen	M	Saudi	22000.00	Certified Physiotherapy Nurse	119
9	123001	Rama	Ali	F	Egyptian	26000.00	Certified Gynecology Nurse	130
10	123002	Joury	Hani	F	Saudi	24000.00	Certified Gynecology Nurse	130
11	123201	Emily	Johnson	F	American	20000.00	Certified Registered Nurse Anesthetist	3208
12	123202	Michael	Garcia	M	American	22000.00	Certified Registered Nurse Anesthetist	3208
13	123701	Ahad	Yousef	F	Saudi	24000.00	Certified Dermatology Nurse	3137
14	123702	Maria	Mazn	F	Saudi	21000.00	Certified Dermatology Nurse	3137
15	124201	Mashal	Hussen	M	Saudi	23000.00	Certified Otolaryngology Nurse	2042
16	124202	Sarah	Saleh	F	Saudi	23000.00	Certified Otolaryngology Nurse	2042
17	124401	Fouz	Ibrahim	F	Saudi	22000.00	Certified Neurology Nurse	6044
18	124402	Selena	Smith	F	American	20000.00	Certified Neurology Nurse	6044
19	124501	Ghadi	Ahmed	F	Saudi	20000.00	Certified Oncology Nurse	3045
20	124502	Albtool	Eyad	F	Saudi	21000.00	Certified Oncology Nurse	3045
21	124601	Mazn	Hmood	M	Saudi	23000.00	Certified Ophthalmic Nurse	46
22	124602	Aram	Abdulaziz	F	Saudi	22000.00	Certified Ophthalmic Nurse	46
23	124801	Draa	Talal	F	Saudi	22000.00	Certified Gastroenterology Nurse	1048
24	124802	Abdelkader	Aimen	M	Saudi	25000.00	Certified Gastroenterology Nurse	1048
25	125501	Loui	Abdelkader	M	Saudi	25000.00	Certified Hematology Nurse	2055
26	125502	Husham	Fars	M	Saudi	26000.00	Certified Hematology Nurse	2055
27	125701	Amal	Ameer	F	Saudi	24000.00	Certified Urology Nurse	4057
28	125702	Hazm	Adeeb	M	Saudi	22000.00	Certified Urology Nurse	4057
29	125801	Sophia	Jad	F	Egyptian	22000.00	Certified Cardiac Nurse	58
30	125802	Nada	Osamah	F	Saudi	21000.00	Certified Cardiac Nurse	58

Staff

INSERT INTO Staff

```
VALUES (311901,'Nora','Abdullah','F','Saudi',11000,'Social Work',08873),
(311902,'Zain','Ibrahiam','M','Saudi',10000,'Social Work',08873),
(311903,'Bader','Rakan','M','Jordanian',10000,'Social Work',08873),
(311904,'Braa','Jad','M','Jordanian',11000,'Social Work',08873),
(311905,'Eyad','Fars','M','Saudi',7800,'Receptionist',08873),
(311906,'Aryam','Abdulkareem','F','Saudi',8000,'Receptionist',08873),
(311907,'Zayd','Mohammed','M','Saudi',8100,'Receptionist',08873),
(311908,'Kadi','Saud','F','Saudi',7500,'Receptionist',08873),
(311909,'Rana','Majad','F','Saudi',7800,'Receptionist',08873),
(311910,'Rahaf','Salem','F','Saudi',8500,'Receptionist',08873),
(311911,'Raghad','Yousef','F','Saudi',7600,'Receptionist',08873),
(311912,'Majed','Saleh','M','Saudi',8300,'Receptionist',08873),
(311913,'Isa','Ahemad','M','Egyptian',7800,'Receptionist',08873),
(311914,'Malik','Abdullah','M','Saudi',7500,'Receptionist',08873),
(311915,'Alanoud','Abdulaziz','F','Saudi',8000,'Receptionist',08873),
(311916,'Musa','Firas','M','Saudi',8100,'Receptionist',08873),
(311917,'Abdulrahman','Saad','M','Saudi',7700,'Receptionist',08873),
(311918,'Refal','Tamim','F','Saudi',8000,'Receptionist',08873),
(311919,'Nawaf','Fasial','M','Saudi',7800,'Receptionist',08873),
(311920,'Turki','Fahad','M','Saudi',18000,'administrative assistant',08873),
(311921,'Thamer','Khalid','M','Saudi',16000,'administrative assistant',08873),
(311922,'Aiman','Hussam','M','Saudi',12000,'Maintenance Technician',18842),
(311923,'Bander','Eyad','M','Saudi',6000,'Security Officer',18842),
(311924,'Abdullah','Yousef','M','Saudi',6000,'Security Officer',18842),
(311925,'Bader','Rakan','M','Saudi',6000,'Security Officer',18842),
(311926,'Yasmeen','Hani','F','Egyptian',9000,'Patient Services Assistants',18842),
(311927,'Farah','Mohannad','F','Jordanian',9000,'Patient Services Assistants',18842),
(311928,'Deem','Bassam','F','Saudi',91000,'Patient Services Assistants',18842),
(311929,'Rami','Mohammed','M','Egyptian',8900,'Patient Services Assistants',18842),
(311930,'Sarah','Johnson','F','American',23000,'HR Manager',38809),
(311931,'Jaber','Fahad','M','Saudi',20000,'HR Coordinator',38809),
(311932,'Fatema','Hatem','F','Saudi',17000,'Recruitment Specialist',38809),
(311933,'Saud','Adeel','M','Saudi',16000,'Training and Development Officer',38809),
(311934,'Dona','Tamim','F','Saudi',20000,'Financial Analyst',28848),
(311935,'Malk','Ibrahiam','F','Saudi',25000,'Accountant',28848),
(311936,'Marwan','Majed','M','Saudi',23000,'Budget Analyst',28848),
(311937,'Wasan','Adeeb','F','Saudi',35000,'Pharmacist',00661),
(311938,'Remaz','Amar','F','Saudi',33000,'Pharmacist',00661),
(311939,'Sultan','Hatem','M','Saudi',36000,'Pharmacist',00661);
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Staff

	StaffID	FirstName	LastName	Gender	Nationality	salary	Role	Dept_ID
1	311901	Nora	Abdullah	F	Saudi	11000.00	Social Work	8873
2	311902	Zain	Ibrahimian	M	Saudi	10000.00	Social Work	8873
3	311903	Bader	Rakan	M	Jordanian	10000.00	Social Work	8873
4	311904	Braa	Jad	M	Jordanian	11000.00	Social Work	8873
5	311905	Eyad	Fars	M	Saudi	7800.00	Receptionist	8873
6	311906	Aryam	Abdulkareem	F	Saudi	8000.00	Receptionist	8873
7	311907	Zayd	Mohanmed	M	Saudi	8100.00	Receptionist	8873
8	311908	Kadi	Saud	F	Saudi	7500.00	Receptionist	8873
9	311909	Rana	Majad	F	Saudi	7800.00	Receptionist	8873
10	311910	Rahaf	Salen	F	Saudi	8500.00	Receptionist	8873
11	311911	Raghad	Yousef	F	Saudi	7600.00	Receptionist	8873
12	311912	Majed	Saleh	M	Saudi	8300.00	Receptionist	8873
13	311913	Isa	Ahemad	M	Egyptian	7800.00	Receptionist	8873
14	311914	Malik	Abdullah	M	Saudi	7500.00	Receptionist	8873
15	311915	Alanoud	Abdulaziz	F	Saudi	8000.00	Receptionist	8873
16	311916	Musa	Firas	M	Saudi	8100.00	Receptionist	8873
17	311917	Abdulrahman	Saad	M	Saudi	7700.00	Receptionist	8873
18	311918	Refal	Tamin	F	Saudi	8000.00	Receptionist	8873
19	311919	Nawaf	Fasial	M	Saudi	7800.00	Receptionist	8873
20	311920	Turki	Fahad	M	Saudi	10000.00	administrative assistant	8873
21	311921	Thamer	Khalid	M	Saudi	16000.00	administrative assistant	8873
22	311922	Aiman	Hussam	M	Saudi	12000.00	Maintenance Technician	18842
23	311923	Bander	Eyad	M	Saudi	6000.00	Security Officer	18842
24	311924	Abdullah	Yousef	M	Saudi	6000.00	Security Officer	18842
25	311925	Bader	Rakan	M	Saudi	6000.00	Security Officer	18842
26	311926	Yasmeen	Hani	F	Egyptian	9000.00	Patient Services Assistants	18842
27	311927	Farah	Mohannad	F	Jordanin	9000.00	Patient Services Assistants	18842
28	311928	Deem	Bassam	F	Saudi	91000.00	Patient Services Assistants	18842
29	311929	Rani	Mohammed	M	Egyptian	8900.00	Patient Services Assistants	18842
30	311930	Sarah	Johnson	F	American	23000.00	HR Manager	38809
31	311931	Jaber	Fahad	M	Saudi	20000.00	HR Coordinator	38809
32	311932	Fatema	Haten	F	Saudi	17000.00	Recruitment Specialist	38809
33	311933	Saud	Adeel	M	Saudi	16000.00	Training and Development Officer	38809
34	311934	Dona	Tamin	F	Saudi	20000.00	Financial Analyst	28848
35	311935	Malk	Ibrahimian	F	Saudi	25000.00	Accountant	28848
36	311936	Marwan	Majed	M	Saudi	23000.00	Budget Analyst	28848
37	311937	Wasan	Adeeb	F	Saudi	35000.00	Pharmacist	661
38	311938	Remaz	Amar	F	Saudi	33000.00	Pharmacist	661
39	311939	Sultan	Haten	M	Saudi	36000.00	Pharmacist	661

MedicalRecord

```
INSERT INTO MedicalRecord
VALUES(9901, 'Fractured arm', 'Apply cast for 6 weeks', 110101, 2221),
(9902, 'Acne', 'Positive progress.Needs Blood Test for Checkup', 110102, 2208),
(9903, 'Anemia', 'Iron supplements and dietary changes', 110103, 2206),
(9904, 'Depression', 'Increase the number of sessions due to negative progress', 110104,
(9905, 'Hypertension', 'Prescribe antihypertensive medication', 110105, 2207),
(9906, 'Appendicitis', 'Surgical removal of appendix', 110106, 2210),
(9907, 'Chronic sinusitis', 'he patient presents with symptoms of chronic sinusitis.
Treatment includes antibiotics, nasal corticosteroids, and saline nasal irrigation.
Follow up in weeks for reassessment', 110107, 2220),
(9908, 'Minor burn', 'Clean and dress wound, prescribe pain relief', 110108, 2209),
(9909, 'LASIK', 'Rest and avoid the light', 110110, 2217),
(9910, 'Birth', ' Prenatal vitamins for breastfeeding support,
Pain medication for postpartum discomfort,
Iron supplements to replenish iron levels after childbirth.', 110110, 2233),
(9911, 'Cows milk allergy', 'More Blood test', 110111, 2211),
(9912, 'Fractured leg', 'Surgery followed by physical therapy', 110112, 2222),
(9913, 'Heart attack', 'Emergency angioplasty and medication', 110113, 2201),
(9914, 'Arthritis', 'Pain management and physical therapy', 110114, 2221),
(9915, 'Kidney stones', 'Pain relief and hydration therapy', 110115, 2238),
(9916, 'Dental checkup', 'Removal of a baby tooth', 110116, 2236),
(9917, 'Overweight', 'Sleeve gastrectomy', 110117, 2214),
(9918, 'Dental checkup', 'Routine examination and cleaning', 110109, 2237),
(9919, 'Alopecia ', ' 3 Cortisone Sessions', 110119, 2208),
(9920, 'Gastroenteritis', 'Fluid replacement and symptomatic treatment', 110120, 2210),
(9921, 'Cavity', 'Tooth filling procedure scheduled for next week.', 110132, 2237),
(9922, 'Dental checkup', 'Routine examination and cleaning', 110128, 2237),
(9923, 'Fractured arm', 'Apply cast for 6 weeks', 110139, 2221),
(9924, 'Dental checkup', 'Routine examination and cleaning', 110130, 2237),
(9925, 'LASIK', 'Rest and avoid the light', 110126, 2218),
(9926, 'LASIK', 'Rest and avoid the light', 110124, 2217),
(9927, 'Anemia', 'Iron supplements and dietary changes', 110122, 2206),
(9928, 'Alopecia ', ' 3 Cortisone Sessions', 110137, 2208),
(9929, 'Eczema', 'Topical corticosteroids and moisturizers.', 110136, 2208),
(9930, 'Celiac Disease', 'Gluten-free diet and nutritional supplements.', 110125, 2211),
(9931, 'Celiac Disease', 'Gluten-free diet and nutritional supplements.', 110127, 2210),
(9932, 'Celiac Disease', 'Gluten-free diet and nutritional supplements.', 110138, 2210);
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MedicalRecordID	Diagnosis	TreatmentDescription	PatientID	DoctorID	
1	9901	Fractured arm	Apply cast for 6 weeks	110101	2221
2	9902	Acne	Positive progress.Needs Blood Test for Checkup	110102	2208
3	9903	Anemia	Iron supplements and dietary changes	110103	2206
4	9904	Depression	Increase the number of sessions due to negative progress	110104	2223
5	9905	Hypertension	Prescribe antihypertensive medication	110105	2207
6	9906	Appendicitis	Surgical removal of appendix	110106	2210
7	9907	Chronic sinusitis	he patient presents with symptoms of chronic sinusitis. Treatment includes antibiotics, nasal corticosteroids, and saline nasal irrigation. Follow up in weeks for reassessment	110107	2220
8	9908	Minor burn	Clean and dress wound, prescribe pain relief	110108	2209
9	9909	LASIK	Rest and avoid the light	110110	2217
10	9910	Birth	Prenatal vitamins for breastfeeding support, Pain medication for postpartum discomfort, Iron supplements to replenish iron levels after childbirth.	110110	2233
11	9911	Cows milk allergy	More Blood test	110111	2211
12	9912	Fractured leg	Surgery followed by physical therapy	110112	2222
13	9913	Heart attack	Emergency angioplasty and medication	110113	2201
14	9914	Arthritis	Pain management and physical therapy	110114	2221
15	9915	Kidney stones	Pain relief and hydration therapy	110115	2238
16	9916	Dental checkup	Removal of a baby tooth	110116	2236
17	9917	Overweight	Sleeve gastrectomy	110117	2214
18	9918	Dental checkup	Routine examination and cleaning	110109	2237
19	9919	Alopecia	3 Cortisone Sessions	110119	2208
20	9920	Gastroenteritis	Fluid replacement and symptomatic treatment	110120	2210
21	9921	Cavity	Tooth filling procedure scheduled for next week.	110132	2237
22	9922	Dental checkup	Routine examination and cleaning	110128	2237
23	9923	Fractured arm	Apply cast for 6 weeks	110139	2221
24	9924	Dental checkup	Routine examination and cleaning	110130	2237
25	9925	LASIK	Rest and avoid the light	110126	2218
26	9926	LASIK	Rest and avoid the light	110124	2217
27	9927	Anemia	Iron supplements and dietary changes	110122	2206
28	9928	Alopecia	3 Cortisone Sessions	110137	2208
29	9929	Eczema	Topical corticosteroids and moisturizers.	110136	2208
30	9930	Celiac Disease	Gluten-free diet and nutritional supplements.	110125	2211
31	9931	Celiac Disease	Gluten-free diet and nutritional supplements.	110127	2210
32	9932	Celiac Disease	Gluten-free diet and nutritional supplements.	110138	2210

Appointments

INSERT INTO Appointments
VALUES

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('2024-01-16', '08:00:00', 2221, 110101, 00119), ('2024-1-20', '10:30:00', 2208, 110102, 03137),  
('2024-03-30', '14:00:00', 2237, 110109, 03111), ('2024-01-10', '09:15:00', 2236, 110116, 03111),  
('2024-02-29', '11:30:00', 2214, 110117, 00203), ('2024-1-20', '13:45:00', 2208, 110119, 03137),  
('2024-02-29', '15:00:00', 2217, 110118, 00046), ('2024-03-30', '08:30:00', 2238, 110115, 06044),  
('2024-01-20', '11:00:00', 2211, 110111, 01048), ('2024-2-22', '14:30:00', 2233, 110110, 00130),  
('2024-03-15', '10:00:00', 2210, 110120, 01048), ('2024-03-16', '10:00:00', 2221, 110139, 00119),  
('2024-06-27', '08:30:00', 2224, 110140, 01202), ('2024-06-27', '9:30:00', 2224, 110119, 01202),  
('2024-06-27', '11:00:00', 2224, 110136, 01202), ('2024-06-27', '9:30:00', 2208, 110133, 03137),  
('2024-06-27', '9:30:00', 2201, 110116, 00058), ('2024-05-17', '13:00:00', 2237, 110132, 03111),  
('2024-05-17', '13:00:00', 2222, 110134, 00119), ('2024-05-17', '14:30:00', 2222, 110131, 00119),  
('2024-05-17', '17:00:00', 2209, 110121, 03137), ('2024-05-17', '14:30:00', 2212, 110129, 06044),  
('2024-06-06', '9:00:00', 2212, 110135, 06044), ('2024-06-06', '17:00:00', 2209, 110123, 03137),  
('2024-06-06', '16:00:00', 2209, 110110, 03137), ('2024-06-06', '9:00:00', 2226, 110113, 00058),  
('2024-03-30', '15:30:00', 2237, 110132, 03111), ('2024-03-30', '12:30:00', 2237, 110128, 03111),  
('2024-03-30', '17:00:00', 2237, 110130, 03111), ('2024-03-30', '12:30:00', 2221, 110114, 00119),  
('2024-02-29', '12:00:00', 2217, 110124, 00046), ('2024-02-29', '12:00:00', 2208, 110137, 03137),  
('2024-02-29', '11:30:00', 2218, 110126, 00046), ('2024-02-29', '11:30:00', 2210, 110138, 01048),  
('2024-3-15', '11:30:00', 2210, 110127, 01048), ('2024-3-15', '10:30:00', 2211, 110125, 01048);
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	AppointmentDate	AppointmentTime	DoctorID	PatientID	Dept_ID
1	1445-06-28	09:15:00	2236	110116	3111
2	1445-07-04	08:00:00	2221	110101	119
3	1445-07-08	10:30:00	2208	110102	3137
4	1445-07-08	11:00:00	2211	110111	1048
5	1445-07-08	13:45:00	2208	110119	3137
6	1445-08-12	14:30:00	2233	110110	130
7	1445-08-19	11:30:00	2210	110138	1048
8	1445-08-19	11:30:00	2214	110117	203
9	1445-08-19	11:30:00	2218	110126	46
10	1445-08-19	12:00:00	2208	110137	3137
11	1445-08-19	12:00:00	2217	110124	46
12	1445-08-19	15:00:00	2217	110110	46
13	1445-09-05	10:00:00	2210	110120	1048
14	1445-09-05	10:30:00	2211	110125	1048
15	1445-09-05	11:30:00	2210	110127	1048
16	1445-09-06	10:00:00	2221	110130	119
17	1445-09-20	08:30:00	2238	110115	6044
18	1445-09-20	12:30:00	2221	110114	119
19	1445-09-20	12:30:00	2237	110120	3111
20	1445-09-20	14:00:00	2237	110109	3111
21	1445-09-20	15:30:00	2237	110132	3111
22	1445-09-20	17:00:00	2237	110130	3111
23	1445-11-09	13:00:00	2222	110134	119
24	1445-11-09	13:00:00	2237	110132	3111
25	1445-11-09	14:30:00	2212	110129	6044
26	1445-11-09	14:30:00	2222	110131	119
27	1445-11-09	17:00:00	2209	110121	3137
28	1445-11-29	09:00:00	2212	110135	6044
29	1445-11-29	09:00:00	2226	110113	58
30	1445-11-29	16:00:00	2209	110120	3137
31	1445-11-29	17:00:00	2209	110123	3137
32	1445-12-21	08:30:00	2224	110140	1202
33	1445-12-21	09:30:00	2201	110116	58
34	1445-12-21	09:30:00	2208	110133	3137
35	1445-12-21	09:30:00	2224	110119	1202
36	1445-12-21	11:00:00	2224	110136	1202

Department

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INSERT INTO Department
VALUES
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```
(03208,'Anesthetics','2021-11-9','Building F32',4),
(08873,'Admissions','2021-11-9','Building F88',1),
(18842,'General Services','2021-11-9','Building F88',2),
(28848,'Finance Department','2021-11-9','Building F88',3),
(38809,'Human Resources','2021-11-9','Building F88',4),
(00058,'Cardiology','2021-11-9','Building F5',1),
(02055,'Hematology','2021-11-9','Building F5',3),
(04057,'Urology','2022-1-5','Building F5',5),
(00130,'Gynecology','2022-5-3','Building F13',1),
(03137,'Dermatology','2023-7-24','Building F13',4),
(00046,'Ophthalmology','2023-10-11','Building F4',1),
(01048,'Gastroenterology','2022-2-5','Building F4',2),
(02042,'Otolaryngology','2024-2-1','Building F4',3),
(03045,'Oncology','2021-12-21','Building F4',4),
(06044,'Neurology','2023-4-3','Building F4',7),
(00203,'Nutrition and Dietetics','2022-10-9','Building F20',1),
(01202,'Psychiatry','2022-6-6','Building F20',2),
(00119,'Physiotherapy','2022-1-5','Building F11',1),
(03111,'Dentistry','2023-7-24','Building F11',4),
(00661,'Pharmacy','2021-11-9','Building F66',0);
```

	Dept_ID	DepartmentName	creationdate	Dept_Location	Dept_Floor
1	46	Ophthalmology	1445-03-26	Building F4	1
2	58	Cardiology	1443-04-04	Building F5	1
3	119	Physiotherapy	1443-06-02	Building F11	1
4	130	Gynecology	1443-10-02	Building F13	1
5	203	Nutrition and Die...	1444-03-13	Building F20	1
6	661	Pharmacy	1443-04-04	Building F66	0
7	1048	Gastroenterology	1443-07-04	Building F4	2
8	1202	Psychiatry	1443-11-07	Building F20	2
9	2042	Otolaryngology	1445-07-20	Building F4	3
10	2055	Hematology	1443-04-04	Building F5	3
11	3045	Oncology	1443-05-17	Building F4	4
12	3111	Dentistry	1445-01-06	Building F11	4
13	3137	Dermatology	1445-01-06	Building F13	4
14	3208	Anesthetics	1443-04-04	Building F32	4
15	4057	Urology	1443-06-02	Building F5	5
16	6044	Neurology	1444-09-12	Building F4	7
17	8873	Admissions	1443-04-04	Building F88	1
18	18842	General Services	1443-04-04	Building F88	2
19	28848	Finance Department	1443-04-04	Building F88	3
20	38809	Human Resources	1443-04-04	Building F88	4

Radiology

```
INSERT INTO Radiology
VALUES (1, 9901, 110101, 'X-ray_fractured_arm.jpg', '2024-01-16'),
(2, 9912, 110112, 'X-ray_fractured_leg.jpg', '2024-01-15'),
(3, 9905, 110105, 'CT_scan_head.jpg', '2024-02-20'),
(4, 9906, 110106, 'X-ray_appendicitis.jpg', '2024-03-10'),
(5, 9907, 110107, 'CT_Scan_Paranasal_Sinuses.pdf', '2024-03-05'),
(6, 9915, 110115, 'Ultrasound_kidney_stones.jpg', '2024-03-30'),
(7, 9923, 110139, 'X-ray_fractured_arm.jpg', '2024-03-16');
```

	ResultID	MedicalRecordID	PatientID	ResultFile	ResultDate
1	1	9901	110101	X-ray_fractured_arm.jpg	1445-07-04
2	2	9912	110112	X-ray_fractured_leg.jpg	1445-07-03
3	3	9905	110105	CT_scan_head.jpg	1445-08-10
4	4	9906	110106	X-ray_appendicitis.jpg	1445-08-29
5	5	9907	110107	CT_Scan_Paranasal_Sinuses.pdf	1445-08-24
6	6	9915	110115	Ultrasound_kidney_stones.jpg	1445-09-20
7	7	9923	110139	X-ray_fractured_arm.jpg	1445-09-06

ClinicalLaboratory

```
INTO ClinicalLaboratory
(4401, 9902, 110102, 'Blood_test_acne.jpg', '2024-01-20'),
9911, 110111, 'Blood_test_infant.jpg', '2024-01-20'),
9917, 110117, 'Blood_test_surgery.jpg', '2024-02-29'),
9910, 110110, 'Blood_Test_Results.pdf', '2024-2-22');
```

	ResultID	MedicalRecordID	PatientID	ResultFile	ResultDate
1	4401	9902	110102	Blood_test_acne.jpg	1445-07-08
2	4402	9911	110111	Blood_test_infant.jpg	1445-07-08
3	4403	9917	110117	Blood_test_surgery.jpg	1445-08-19
4	4404	9910	110110	Blood_Test_Results.pdf	1445-08-12

PatientFeedback

```
INSERT INTO PatientFeedback
VALUES (7701, 110101, 5, 'No Suggestions', '2024-01-18'),
(7702, 110102, 5, 'No Suggestions.', '2024-1-25'),
(7703, 110107, 3, 'Waiting time was a bit long, but overall satisfied with treatment.', '2024-03-08'),
(7704, 110114, 4, 'No Suggestions.', '2024-04-10'),
(7705, 110118, 2, 'Would appreciate more detailed explanations from the doctor.', '2024-02-29'),
(7706, 110119, 5, 'No Suggestions', '2024-01-20'),
(7707, 110113, 4, 'Improved signage needed within the hospital.', '2024-01-27'),
(7708, 110115, 3, 'Would like more comfortable waiting areas.', '2024-03-30'),
(7719, 110106, 4, 'No Suggestions', '2024-03-10'),
(7710, 110120, 2, 'Suggest providing more amenities for chemotherapy patients.', '2024-01-17');
```

	FeedbackID	▼	PatientID	▼	SatisfactionRate	▼	Suggestions	▼	FeedbackDate
1	7701		110101		5		No Suggestions		1445-07-06
2	7702		110102		5		No Suggestions.		1445-07-13
3	7703		110107		3		Waiting time was a bit long, but overall satisfied with treatment.		1445-08-27
4	7704		110114		4		No Suggestions.		1445-10-01
5	7705		110118		2		Would appreciate more detailed explanations from the doctor.		1445-08-19
6	7706		110119		5		No Suggestions		1445-07-08
7	7707		110113		4		Improved signage needed within the hospital.		1445-07-15
8	7708		110115		3		Would like more comfortable waiting areas.		1445-09-20
9	7710		110120		2		Suggest providing more amenities for chemotherapy patients.		1445-07-05
10	7719		110106		4		No Suggestions		1445-08-29

Pharmacy

INSERT INTO Pharmacy
VALUES

```
(1001, 110101, 000.00, 2221, 'Painkillers and antibiotics', '2024-01-16'),  
(1002, 110102, 000.00, 2208, 'Accutane', '2024-1-20'),  
(1003, 110103, 000.00, 2206, 'Iron supplements', '2024-03-05'),  
(1004, 110114, 000.00, 2221, 'Pain relief ointment and medication', '2024-04-10'),  
(1005, 110118, 50.00, 2217, 'Eyes drop', '2024-02-29'),  
(1006, 110111, 00.00, 2211, 'Neocate LCP', '2024-01-20'),  
(1007, 110119, 00.00, 2208, 'Regaine', '2024-1-20'),  
(1008, 110104, 100.00, 2223, 'antidepressants', '2024-08-30'),  
(1009, 110109, 000.00, 2237, 'Dental hygiene products', '2024-03-30'),  
(1010, 110106, 00.00, 2210, 'Pain relief cream ', '2024-03-10'),  
(1011, 110112, 90.00, 2222, 'Pain medicationn', '2024-11-15');
```

	BillNumber	PatientID	Cost	DoctorID	Prescription	PrescriptionDate
1	1001	110101	0.00	2221	Painkillers and antibiotics	1445-07-04
2	1002	110102	0.00	2208	Accutane	1445-07-08
3	1003	110103	0.00	2206	Iron supplements	1445-08-24
4	1004	110114	0.00	2221	Pain relief ointment and medication	1445-10-01
5	1005	110118	50.00	2217	Eyes drop	1445-08-19
6	1006	110111	0.00	2211	Neocate LCP	1445-07-08
7	1007	110119	0.00	2208	Regaine	1445-07-08
8	1008	110104	100.00	2223	antidepressants	1446-02-26
9	1009	110109	0.00	2237	Dental hygiene products	1445-09-20
10	1010	110106	0.00	2210	Pain relief cream	1445-08-29
11	1011	110112	90.00	2222	Pain medicationn	1446-05-13

DDL (Views)

1

How many uninsured patients are there by gender?

```
CREATE VIEW no_insurance AS  
SELECT p.Gender, COUNT(p.Gender) AS Number_Gender  
FROM Patient p  
WHERE p.INSURANCENUMBER = 'No Insurance'  
GROUP BY p.Gender;
```

Results Messages

Gender	Number_Gender
F	9
M	9

2

What clinical laboratory results are available for patients, including names, contact information, result dates, and files?"

```
CREATE VIEW patients_ClinicalLaboratory AS  
SELECT Patient.PatientID, Patient.FirstName, Patient.LastName, Patient.PhoneNumber, ClinicalLaboratory.RESULTFILE, ClinicalLaboratory.RESULTDATE  
FROM Patient INNER JOIN ClinicalLaboratory ON Patient.PatientID = ClinicalLaboratory.PatientID;
```

Messages

PatientID	FirstName	LastName	PhoneNumber	RESULTFILE	RESULTDATE
10102	Lamar	Fahad	0504599344	Blood_test_acne.jpg	1445-07-08
10111	Firas	Fasial	0505446443	Blood_test_infant.jpg	1445-07-08
10117	Yousef	Zaid	0501804638	Blood_test_surgery.jpg	1445-08-19
0110	Bayan	Salman	0505446443	Blood_Test_Results.pdf	1445-08-12

3

What is the information about the specialized doctors in the Department of Cardiology?

```
CREATE VIEW Cardiology_department AS
SELECT *
FROM Doctor
WHERE DEPT_ID = 58
with check option;
```

Messages

DoctorID	FirstName	LastName	Gender	Nationality	salary	Certification	Qualification	Dept_ID
2201	Diana	Geisser	F	American	40000.00	MD,ABIM	Cardiologist	58
2202	Majed	Yaseer	M	Saudi	30000.00	MD	Paediatric Cardiologist	58
2203	Tala	Ahmad	F	Saudi	30000.00	MD	Cardiologist	58
2225	John	Hutson	M	British	45000.00	MD	Surgeon	58
2226	Juan	Bertagary	M	British	45000.00	MD	Surgeon	58

```
INSERT INTO Cardiology_department VALUES (2250,'Mona','Geisser','F','American',45000,'MD,ABIM','Cardiologist',04050);
INSERT INTO Cardiology_department VALUES (2245,'Saad','Turki','M','Saudi',33000,'MD,ABA','Consultant Anaesthetist',03200);
```

3-- How many prescriptions and details are available for uninsured patients?

Messages

6:00 PM Started executing query at Line 402
(1 row affected)
Msg 504, Level 16, State 1, Line 4
The attempted insert or update failed because the target view either specifies WITH CHECK OPTION or spans a view that specifies WITH CHECK OPTION and the statement did not qualify under the CHECK OPTION constraint.
The statement has been terminated.
Total execution time: 00:00:00.017

after
insert

```
INSERT INTO Cardiology_department VALUES (2250,'Mona','Geisser','F','American',45000,'MD,ABIM','Cardiologist',04050);
INSERT INTO Cardiology_department VALUES (2245,'Saad','Turki','M','Saudi',33000,'MD,ABA','Consultant Anaesthetist',03200);
```

3-- How many prescriptions and details are available for uninsured patients?

Messages

DoctorID	FirstName	LastName	Gender	Nationality	salary	Certification	Qualification	Dept_ID
2201	Diana	Geisser	F	American	40000.00	MD,ABIM	Cardiologist	58
2202	Majed	Yaseer	M	Saudi	30000.00	MD	Paediatric Cardiologist	58
2203	Tala	Ahmad	F	Saudi	30000.00	MD	Cardiologist	58
2225	John	Hutson	M	British	45000.00	MD	Surgeon	58
2226	Juan	Bertagary	M	British	45000.00	MD	Surgeon	58
2250	Mona	Geisser	F	American	45000.00	MD,ABIM	Cardiologist	58

4

What prescriptions and details are available for uninsured patients?

```
CREATE VIEW prescription noinsurance AS
SELECT Pharmacy.BillNumber, Patient.PatientID, Patient.FirstName, Pharmacy.prescription
FROM Patient JOIN Pharmacy on Patient.PatientID = Pharmacy.PatientID
WHERE Patient.InsuranceNumber = 'No Insurance';
```

Results Messages

BillNumber	PatientID	FirstName	prescription
1005	110118	Manar	Eyes drop
1008	110104	Leen	antidepressants
1011	110112	Abdullah	Pain medicationn

5

What patients received a low satisfaction rating?

```
CREATE VIEW bad_feedback AS  
SELECT patient.FirstName, patient.LastName  
FROM Patient JOIN patientfeedback on Patient.PatientID =PatientFeedback.PatientID  
WHERE SatisfactionRate <=2;
```

Results Messages

FirstName	LastName
Manar	Rayan
Marah	Saad

DML (Queries)

1

How many patients had appointments on February 22, 2024, in the Gynecology Department?

```
SELECT COUNT(*) AS NumberOfPatients
FROM Appointments NATURAL JOIN Department
WHERE Department.DepartmentName = 'Gynecology'
AND AppointmentDate = '2024-2-22';
```

	NumberOfPatients	▼
1	1	

2

Which patients are diagnosed with Overweight?

```
SELECT FirstName, LastName
FROM Patient
WHERE PatientID IN (
SELECT PatientID
FROM MedicalRecord
WHERE Diagnosis = 'Overweight');
```

Results Messages

	FirstName	▼	LastName	▼
1	Yousef		Zaid	

3

What are the names of male doctors working on Floor 1 ?

```
SELECT FirstName, LastName
FROM Doctor WHERE Gender = 'M' AND Dept_ID IN
(SELECT Dept_ID FROM Department WHERE Dept_Floor = 1);
```

ts Messages

FirstName	LastName
Majed	Yaseer
Saleh	Naif
Baseel	Tariq
John	Hutson
Juan	Bortagaray

4

Which patients have a satisfaction rating of less than 4, and what are their suggestions?

```
SELECT PF.PatientID, P.FirstName, SatisfactionRate, PF.Suggestions
FROM PatientFeedback PF
JOIN Patient P ON PF.PatientID = P.PatientID
WHERE PF.SatisfactionRate < 4;
```

Results Messages

	PatientID	FirstName	SatisfactionRate	Suggestions
1	110107	Hussam	3	Waiting time was a bit long, but overall satisfied w...
2	110118	Manar	2	Would appreciate more detailed explanations from the...
3	110115	Nasser	3	Would like more comfortable waiting areas.
4	110120	Marah	2	Suggest providing more amenities for chemotherapy pa...

5

How many patients have a
Strawberries allergy?

```
SELECT COUNT(*) AS NumOfPatientsWithStrawberriesAllergy
FROM Patient
WHERE Allergies = 'Strawberries'
```

Results Messages

NumOfPatientsWithStrawberriesAllergy
2

6

Which male patients don't
suffer from allergies?

```
SELECT Patient.PatientID, Patient.FirstName, Patient.LastName
FROM Patient
WHERE Patient.Allergies LIKE 'No Allergies' and Gender = 'M';
```

Results Messages

	PatientID	FirstName	LastName
1	110105	Sultan	Mohammed
2	110111	Firas	Fasial
3	110113	Saleh	Ahemad
4	110119	Omar	Osamah
5	110131	Yousef	Majed
6	110134	Saud	Fars
7	110136	Marvan	Malk

7

What are the radiological reports for patient ID 110107?

```
SELECT R.ResultFile, R.ResultDate
FROM Radiology R JOIN MedicalRecord MR ON R.MedicalRecordID = MR.MedicalRecordID
WHERE MR.PatientID = 110107;
```

Results Messages

	ResultFile	ResultDate
1	CT_Scan_Paranasal_Sinuses.pdf	1445-08-24

8

Who are the female reception staff members with salaries higher than \$8000?

```
SELECT FirstName, LastName , salary
FROM Staff
WHERE Role = 'Receptionist'
AND Gender = 'F' AND salary > 8000;
```

Results Messages

	FirstName	LastName	salary
1	Rahaf	Salem	8500.00

9

Which patients have only one appointment in the 'Neurology' department?

```
SELECT P.PatientID, P.FirstName, P.LastName
FROM Patient P , Appointments A , Department Dept
WHERE P.PatientID = A.PatientID AND A.Dept_ID = Dept.Dept_ID
AND Dept.DepartmentName = 'Neurology'
GROUP BY P.PatientID , P.FirstName, P.LastName
HAVING COUNT(*) = 1;
```

Results Messages

	PatientID	FirstName	LastName
1	110115	Nasser	Saleh
2	110129	Asmaa	Omar
3	110135	Farah	Isa

10

Which patients rated poorly (below 3)?

```
SELECT P.PatientID, P.FirstName, P.LastName, PF.SatisfactionRate
FROM Patient P
JOIN PatientFeedback PF ON P.PatientID = PF.PatientID
WHERE PF.SatisfactionRate < 3;
```

Results Messages

	PatientID	FirstName	LastName	SatisfactionRate
1	110118	Manar	Rayan	2
2	110120	Marah	Saad	2

11

How should the salary of administrative assistants be updated if their salary is less than \$18,000, by increasing it by \$1,000?

```
UPDATE Staff
SET salary = salary + 1000
WHERE Role = 'administrative assistant' AND salary < 18000;

Select FirstName, LastName, salary
From staff
Where Role = 'administrative assistant'
and salary <= 18000
```

Results Messages

FirstName	LastName	salary
Turki	Fahad	18000.00
Thamer	Khalid	17000.00

12

Find the female nurses whose salary is less than 22,000 and increase their salaries by 1000?

```
UPDATE Nurse
SET salary = salary + 1000
WHERE Gender = 'F' AND salary < 22000;
```

Messages

4:26:01 PM Started executing query at Line 734
(8 rows affected)
Total execution time: 00:00:00.027

```
Select FirstName, LastName, salary
FROM Nurse
WHERE Gender = 'F'

Select FirstName, LastName, salary
FROM Nurse
WHERE Gender = 'F'
AND salary <= 22000;
```

Results Messages

FirstName	LastName	salary
Ariana	Smith	22000.00
Emma	Gomez	22000.00
Maria	Chang	22000.00
Emily	Johnson	21000.00
Maria	Mazin	22000.00
Fouz	Ibrahim	22000.00
Selena	Smith	21000.00
Ghadi	Ahmed	21000.00
Albtool	Eyad	22000.00
Aram	Abdulaziz	22000.00
Draa	Talal	22000.00
Sophia	Jad	22000.00
Nada	Osamah	22000.00

13

What are the Nurse ID, first name, and last name of male nurses who are of Saudi nationality, ordered by first name?

```
select NurseID,FirstName,LastName
from Nurse
where Nationality ='saudi' and Gender like 'M'
order by FirstName;
```

Results		Messages		
	NurseID	FirstName	LastName	
1	124802	Abdelkader	Aimen	
2	125702	Hazm	Adeeb	
3	121901	Hazza	Faisal	
4	121902	Husham	Hussen	
5	125502	Husham	Fars	
6	125501	Loui	Abdelkader	
7	124201	Mashal	Hussen	
8	124601	Mazn	Hmood	

14

Which patients have upcoming appointments and what are their medical records?

```
SELECT Patient.PatientID, Patient.FirstName, Patient.LastName, Patient.PhoneNumber, Appointments.AppointmentDate, MedicalRecord.Diagnosis
FROM Patient INNER JOIN Appointments ON Patient.PatientID = Appointments.PatientID, MedicalRecord
WHERE Appointments.AppointmentDate >= GETDATE() and Patient.PatientID = MedicalRecord.PatientID;
```

Results		Messages				
	PatientID	FirstName	LastName	PhoneNumber	AppointmentDate	Diagnosis
1	110110	Bayan	Salman	0505146443	1445-11-29	Birth
2	110113	Saleh	Ahemad	0504894258	1445-11-29	Heart attack
3	110116	Loena	Ahemad	0538015577	1445-12-21	Dental checkup
4	110119	Omar	Osomah	0537605745	1445-12-21	Alopecia
5	110132	Maha	Ibrahim	0257632211	1445-11-09	Cavity
6	110156	Marwan	Malk	0504873334	1445-12-21	Eczema

15

What is the department name, gender, and average salary for staff members with salaries greater than or equal to \$5000, grouped by department name and gender, and ordered accordingly?

```
SELECT d.DepartmentName, s.Gender, AVG(s.salary) AS AverageSalary
FROM Staff s
RIGHT JOIN Department d ON s.Dept_ID = d.Dept_ID
WHERE s.salary >= 5000
GROUP BY d.DepartmentName, s.Gender
ORDER BY d.DepartmentName, s.Gender;
```

Results		Messages	
	DepartmentName	Gender	AverageSalary
1	Admissions	F	8300.000000
2	Admissions	M	9853.846153
3	Finance Department	F	22500.000000
4	Finance Department	M	23000.000000
5	General Services	F	36333.333333
6	General Services	M	7780.000000
7	Human Resources	F	20000.000000
8	Human Resources	M	18000.000000
9	Pharmacy	F	34000.000000
10	Pharmacy	M	36000.000000

16

What is the maximum salary for each department among both nurses and staff members?

```
SELECT DepartmentName, MAX(salary) AS MaxSalary
FROM (
  SELECT Dept_ID, salary FROM Nurse
  UNION ALL
  SELECT Dept_ID, salary FROM Staff
) AS AllSalaries
JOIN Department ON AllSalaries.Dept_ID = Department.Dept_ID
GROUP BY DepartmentName;
```

Results	Messages
DepartmentName	MaxSalary
1 Admissions	18000.00
2 Anesthetics	22000.00
3 Cardiology	22000.00
4 Dentistry	23000.00
5 Dermatology	24000.00
6 Finance Department	25000.00
7 Gastroenterology	25000.00
8 General Services	91000.00
9 Gynecology	20000.00
10 Hematology	26000.00
11 Human Resources	23000.00
12 Neurology	22000.00
13 Nutrition and Dietetics	23000.00
14 Oncology	21000.00
15 Ophthalmology	23000.00
16 Otolaryngology	23000.00
17 Pharmacy	36000.00
18 Physiotherapy	22000.00
19 Psychiatry	23000.00
20 Urology	24000.00

17

What are the maximum and minimum salaries for each nationality among staff members?

```
SELECT Nationality, MAX(salary) AS MaxSalary, MIN(salary) AS MinSalary
FROM Staff
GROUP BY Nationality;
```

Results Messages

	Nationality	MaxSalary	MinSalary
1	American	23000.00	23000.00
2	Egyptian	9000.00	7800.00
3	Jordanian	11000.00	10000.00
4	Jordanin	9000.00	9000.00
5	Saudi	91000.00	6000.00

18

What are the maximum and minimum salaries for each role among male staff members?

```
SELECT Role, MAX(salary) AS MaxSalary, MIN(salary) AS MinSalary
FROM Staff
WHERE Gender = 'M'
GROUP BY Role;
```

Results Messages

	Role	MaxSalary	MinSalary
1	administrative assistant	18000.00	16000.00
2	Budget Analyst	23000.00	23000.00
3	HR Coordinator	20000.00	20000.00
4	Maintenance Technician	12000.00	12000.00
5	Patient Services Assistants	8900.00	8900.00
6	Pharmacist	36000.00	36000.00
7	Receptionist	8300.00	7500.00
8	Security Officer	6000.00	6000.00
9	Social Work	11000.00	10000.00
10	Training and Development Officer	16000.00	16000.00

19

What is the average cost of prescriptions for each gender?

```
SELECT pa.Gender, AVG(p.Cost) AS AverageCost
FROM Pharmacy p
JOIN Patient pa ON p.PatientID = pa.PatientID
GROUP BY pa.Gender;
```

Results Messages

	Gender	AverageCost
1	F	21.428571
2	M	22.500000

20

Which department has the most female staff members, including doctors and nurses?

```
SELECT d.DepartmentName, COUNT(*) AS FemaleStaffCount
FROM (
  SELECT Dept_ID, Gender FROM Staff
  UNION ALL
  SELECT Dept_ID, Gender FROM Doctor
  UNION ALL
  SELECT Dept_ID, Gender FROM Nurse
) AS CombinedData
JOIN Department d ON CombinedData.Dept_ID = d.Dept_ID
WHERE CombinedData.Gender = 'F'
GROUP BY d.DepartmentName
ORDER BY FemaleStaffCount DESC;
```

Results Messages

	DepartmentName	FemaleStaffCount
1	Admissions	0
2	Dermatology	5
3	Neurology	5
4	Nutrition and Dietetics	4
5	Gynecology	4
6	Cardiology	4
7	Oncology	3
8	General Services	3
9	Ophthalmology	2
10	Otolaryngology	2
11	Pharmacy	2
12	Physiotherapy	2
13	Psychiatry	2
14	Urology	2
15	Dentistry	2
16	Hematology	2
17	Human Resources	2
18	Finance Department	2
19	Gastroenterology	2
20	Anesthetics	1

21

Show me the prescription for patients with insurance, order by their id

```
SELECT ph.Prescription, pa.InsuranceNumber, ph.patientID
FROM Pharmacy ph
JOIN Patient pa ON ph.PatientID = pa.PatientID
WHERE pa.InsuranceNumber not like 'NO Insurance%'
ORDER BY pa.PatientID;
```

Results Messages

	Prescription	InsuranceNumber	patientID
1	Painkillers and antibiotics	111-021-001	110101
2	Accutane	112-022-221	110102
3	Iron supplements	111-221,920	110103
4	Pain relief cream	112-989-030	110106
5	Dental hygiene products	221-221-021	110109
6	Neocate LCP	224-443-343	110111
7	Pain relief ointment and medication	110-903-321	110114
8	Regaine	506-911-120	110119

22

What is the average age of people who have allergies, categorized by allergy status?

```
SELECT AllergyStatus, AVG(Age) AS AverageAge
FROM (
  SELECT
    CASE WHEN Allergies NOT LIKE 'No%' THEN 'With Allergies' ELSE 'Without Allergies' END AS AllergyStatus,
    Age
  FROM Patient
) AS SubQuery
WHERE AllergyStatus = 'With Allergies'
GROUP BY AllergyStatus;
```

Results Messages

AllergyStatus	AverageAge
With Allergies	21

23

What is the percentage of male and female patients in each blood type category?

```
SELECT BloodType, Gender,
COUNT(*) AS TotalPatients,
(COUNT(*) * 100.0) / SUM(COUNT(*)) OVER(PARTITION BY BloodType) AS PercentageByBloodType
FROM Patient
GROUP BY BloodType, Gender;
```

	BloodType	Gender	TotalPatients	PercentageByBloodType
1	A+	F	2	25.000000000000000
2	A+	M	6	75.000000000000000
3	A-	F	1	50.000000000000000
4	A-	M	1	50.000000000000000
5	AB+	F	4	57.142857142857142
6	AB+	M	3	42.857142857142857
7	B+	F	3	30.000000000000000
8	B+	M	7	70.000000000000000
9	B-	F	1	50.000000000000000
10	B-	M	1	50.000000000000000
11	O+	F	8	72.727272727272727
12	O+	M	3	27.272727272727272

24

How many staff members, including doctors and nurses, are there in each department categorized by gender, and what are the results ordered by department name?

```
SELECT d.DepartmentName, AllStaff.Gender, COUNT(*) AS TotalStaff
FROM: SELECT Dept_ID, Gender FROM Staff
UNION ALL
SELECT Dept_ID, Gender FROM Doctor
UNION ALL
SELECT Dept_ID, Gender FROM Nurse t as AllStaff
JOIN Department d ON AllStaff.Dept_ID = d.Dept_ID
GROUP BY d.DepartmentName, AllStaff.Gender
ORDER BY d.DepartmentName;
```

	DepartmentName	Gender	TotalStaff
1	Admissions	F	8
2	Admissions	M	13
3	Anesthetics	F	1
4	Anesthetics	M	3
5	Cardiology	F	4
6	Cardiology	M	1
7	Dentistry	F	2
8	Dentistry	M	4
9	Dermatology	F	5
10	Dermatology	M	1
11	Finance Department	F	2
12	Finance Department	M	1
13	Gastroenterology	F	2
14	Gastroenterology	M	2
15	General Services	F	3
16	General Services	M	5
17	Gynecology	F	4
18	Hematology	F	2
19	Hematology	M	2
20	Human Resources	F	2
21	Human Resources	M	2
22	Neurology	F	5
23	Neurology	M	1
24	Nutrition and Diet	F	4
25	Oncology	F	3
26	Oncology	M	1
27	Ophthalmology	F	2
28	Ophthalmology	M	3
29	Otolaryngology	F	2
30	Otolaryngology	M	2
31	Pharmacy	F	2
32	Pharmacy	M	1
33	Physiotherapy	F	2
34	Physiotherapy	M	2
35	Psychiatry	F	2
36	Psychiatry	M	2
37	Urology	F	2
38	Urology	M	1

25

What is the department name and the number of nurses working in each department?

```
SELECT d.DepartmentName, COUNT(n.NurseID) AS NumberOfNurses
FROM Nurse n
INNER JOIN Department d ON n.Dept_ID = d.Dept_ID
GROUP BY d.DepartmentName;
```

	DepartmentName	NumberOfNurses
1	Anesthetics	2
2	Cardiology	2
3	Dentistry	2
4	Dermatology	2
5	Gastroenterology	2
6	Gynecology	2
7	Hematology	2
8	Neurology	2
9	Nutrition and Dietetics	2
10	Oncology	2
11	Ophthalmology	2
12	Otolaryngology	2
13	Physiotherapy	2
14	Psychiatry	2
15	Urology	2

26

What patients have upcoming appointments and what is their contact information?

```
SELECT Patient.PatientID, Patient.FirstName, Patient.LastName, Patient.PhoneNumber, Appointments.AppointmentDate
FROM Patient
INNER JOIN Appointments ON Patient.PatientID = Appointments.PatientID
WHERE Appointments.AppointmentDate >= GETDATE();
```

	PatientID	FirstName	LastName	PhoneNumber	AppointmentDate
1	110134	Saad	Farrs	0503054900	1445-11-09
2	110132	Maha	Ibrahim	0557632211	1445-11-09
3	110120	Aasma	Omar	0553801275	1445-11-00
4	110131	Yusef	Majed	0509712704	1445-11-09
5	110121	Haya	Rasheed	05074319034	1445-11-09
6	110135	Farah	Tua	0505990887	1445-11-20
7	110113	Saleh	Ahmed	0504094250	1445-11-29
8	110110	Bayan	Salaan	0505444443	1445-11-29
9	110123	Lana	Sultan	0505587433	1445-11-20
10	110140	Saad	Malik	0509078432	1445-12-21
11	110116	Leena	Ahmed	0538011577	1445-12-21
12	110133	Khalid	Abdelrahman	0505067433	1445-12-21
13	110119	Omar	Oyadah	0537061745	1445-12-21
14	110136	Muhammad	Malik	0509071234	1445-12-21

27

What are the top 5 salaries for doctors excluding those in the Urology department?

```
SELECT top 5*
FROM Doctor
WHERE Dept_ID != (SELECT Dept_ID
FROM Department
WHERE DepartmentName = 'Urology')
ORDER BY salary DESC ;
```

Results	Messages								
	DoctorID	FirstName	LastName	Gender	Nationality	salary	Certification	Qualification	Dept_ID
1	2284	George	Chalkiadis	M	Saudi	60000.00	MD,BLS	Anesthesiologist	3208
2	2288	Marah	Abdulkarim	F	Saudi	60000.00	MD,ABD	Dermatologists	3137
3	2285	Saul	Turki	M	Saudi	50000.00	MD,ABA	Consultant Anaesthetist	3208
4	2238	Mohammed	Abdelrahman	M	Saudi	50000.00	MD	Nephrologist	6044
5	2239	Ling	Chen	F	Chinese	50000.00	MD	Nephrologist	6044

28

Who is the patient tested in the lab on February 29, 2024, and what are the details of the test?

```
SELECT CL.ResultID, CL.MedicalRecordID, CL.PatientID, P.FirstName,P.LastName, CL.ResultFile, CL.ResultDate
FROM ClinicalLaboratory CL, Patient P
WHERE CL.PatientID = P.PatientID AND CL.ResultDate = '2024-02-29';
```

s Messages

ResultID	MedicalRecordID	PatientID	FirstName	LastName	ResultFile	ResultDate
1403	9917	110117	Yousef	Zaid	Blood_test_surgery.jpg	1445-08-19

29

What are the names of doctors, along with their appointment counts and status, for appointments scheduled on March 30, 2024, between 12:00 PM and 5:59 PM?

```
SELECT D.FirstName, D.LastName, AppointmentCount,
CASE
    WHEN AppointmentCount > 3 THEN 'BUSY'
    ELSE 'available'
END AS DoctorStatus
FROM (
    SELECT DoctorID, AppointmentDate, COUNT(*) AS AppointmentCount
    FROM Appointments
    WHERE AppointmentDate = '2024-03-30' AND ( AppointmentTime LIKE '12%' OR AppointmentTime LIKE '13%' OR AppointmentTime LIKE '14%' OR AppointmentTime LIKE '15%' OR
    AppointmentTime LIKE '16%' OR AppointmentTime LIKE '17%')
    GROUP BY DoctorID, AppointmentDate
) AS p
INNER JOIN Doctor D ON p.DoctorID = D.DoctorID;
```

Messages			
FirstName	LastName	AppointmentCount	DoctorStatus
Rada	Saleh	1	available
Mohraf	Hamad	4	BUSY

30

How are staff members categorized into salary categories (“Low-Paid,” “Medium-Paid,” or “High-Paid”) based on their average salaries, grouped by role and gender, and ordered by ascending average salary?

```
SELECT
CASE
    WHEN AVG(Salary) <= 20000 THEN 'Low-Paid'
    WHEN AVG(Salary) > 20000 AND AVG(Salary) <= 30000 THEN 'Medium-Paid'
    ELSE 'High-Paid'
END AS SalaryCategory,
Role,
Gender,
ROUND(AVG(Salary), 2) AS AvgSalary
FROM Staff
GROUP BY Role, Gender
ORDER BY AvgSalary ASC;
```

Results		Messages		
	SalaryCategory	Role	Gender	AvgSalary
3	Low-Paid	Security Officer	M	6666.666666
2	Low-Paid	Receptionist	M	7887.500000
2	Low-Paid	Receptionist	F	7914.200000
4	Low-Paid	Patient Services Assistants	M	8966.000000
5	Low-Paid	Social Work	M	18333.330000
6	Low-Paid	Social Work	F	11000.000000
7	Low-Paid	Maintenance Technician	M	12000.000000
7	Low-Paid	Training and Development Officer	M	16000.000000
8	Low-Paid	Recruitment Specialist	F	17000.000000
10	Low-Paid	administrative assistant	M	18000.000000
11	Low-Paid	Financial Analyst	F	20000.000000
12	Low-Paid	HR coordinator	M	20000.000000
13	Medium-Paid	HR Manager	F	23000.000000
14	Medium-Paid	Budget Analyst	M	23000.000000
15	Medium-Paid	Accountant	F	25000.000000
20	High-Paid	Pharmacist	F	34000.000000
17	High-Paid	Pharmacist	M	36000.000000
10	High-Paid	Patient Services Assistants	F	36333.330000

31

How many patients have celiac disease, and what are their blood type and age range?

```
SELECT p.PatientID, p.BloodType,  
CASE WHEN p.BloodType LIKE '%+' THEN '+'  
WHEN p.BloodType LIKE '%-' THEN '-'  
ELSE 'Unknown' END AS BloodTypeRh,  
CASE WHEN p.age < 18 THEN 'Under 18'  
WHEN p.age BETWEEN 18 AND 30 THEN '18-30'  
WHEN p.age BETWEEN 31 AND 50 THEN '31-50'  
ELSE 'Over 50' END AS AgeRange  
FROM Patient p  
JOIN MedicalRecord mr ON p.PatientID = mr.PatientID  
WHERE mr.Diagnosis = 'Celiac Disease';
```

Results Messages

	PatientID	BloodType	BloodTypeRh	AgeRange
1	110125	A-	-	31-50
2	110127	A+	+	31-50
3	110138	O+	+	Over 50

32

What appointments are scheduled for the Dentistry department?

```
SELECT *  
FROM Appointments  
WHERE Dept_ID = (  
SELECT Dept_ID  
FROM Department  
WHERE DepartmentName LIKE 'Dentistry'  
);
```

Results Messages

	AppointmentDate	AppointmentTime	DoctorID	PatientID	Dept_ID
1	1445-06-28	09:15:00	2236	110116	3111
2	1445-09-20	12:30:00	2237	110128	3111
3	1445-09-20	14:00:00	2237	110109	3111
4	1445-09-20	15:30:00	2237	110132	3111
5	1445-09-20	17:00:00	2237	110130	3111
6	1445-11-09	13:00:00	2237	110132	3111

33

What is the renewal status of each department based on their creation date?(if the department was opened before 2022-06-1 then it needs a renewal)

```
SELECT DepartmentName,
CASE
WHEN creationdate < '2022-06-01' THEN 'Needs Renewal'
ELSE 'Does Not Need Renewal'
END AS RenewalStatus
FROM Department;
```

Results	Messages
DepartmentName	RenewalStatus
1 Ophthalmology	Does Not Need Renewal
2 Cardiology	Needs Renewal
3 Physiotherapy	Needs Renewal
4 Gynecology	Needs Renewal
5 Nutrition and Dietetics	Does Not Need Renewal
6 Pharmacy	Needs Renewal
7 Gastroenterology	Needs Renewal
8 Psychiatry	Does Not Need Renewal
9 Otolaryngology	Does Not Need Renewal
10 Hematology	Needs Renewal
11 Oncology	Needs Renewal
12 Dentistry	Does Not Need Renewal
13 Dermatology	Does Not Need Renewal
14 Anesthetics	Needs Renewal
15 Urology	Needs Renewal
16 Neurology	Does Not Need Renewal
17 Admissions	Needs Renewal
18 General Services	Needs Renewal
19 Finance Department	Needs Renewal
20 Human Resources	Needs Renewal

34

Which clinic has the most appointments?

```
SELECT TOP 1 DepartmentName, AppointmentCount
FROM (
SELECT d.DepartmentName, COUNT(*) AS AppointmentCount
FROM Appointments a
JOIN Department d ON a.Dept_ID = d.Dept_ID
GROUP BY d.DepartmentName
) AS DepartmentAppointments
ORDER BY AppointmentCount DESC;
```

Results	Messages
DepartmentName	AppointmentCount
1 Dermatology	7

35

Add a new attribute called
yearsOfService to the table nurse
and take only integers

FOR TABLE Nurse
ADD YearsOfService INT;
select * from Nurse

NurseID	FirstName	LastName	Gender	Nationality	salary	Certification	Dept_ID	YearsOfService
120201	Sophia	Nhan	F	Pakistani	21000.00	Certified Psychiatric Nurse	1202	AVLE
120202	Arlene	Smith	F	American	21000.00	Certified Psychiatric Nurse	1202	AVLE
120301	Emma	Gomez	F	American	21000.00	Certified Nutrition and Dietetics Nurse	203	AVLE
120302	Emily	Johnson	F	American	21000.00	Certified Nutrition and Dietetics Nurse	203	AVLE
121101	Maria	Chang	F	Chinese	22000.00	Certified Dental Nurse	3111	AVLE
121102	Michael	Nguyen	M	Vietnamese	21000.00	Certified Dental Nurse	3111	AVLE
121301	Mazza	Faisal	M	Saudi	21000.00	Certified Physiotherapy Nurse	119	AVLE
121302	Muham	Hussen	M	Saudi	22000.00	Certified Physiotherapy Nurse	119	AVLE
122001	Rana	Ali	F	Egyptian	16000.00	Certified Gynecology Nurse	110	AVLE
122002	Jocry	Hani	F	Saudi	14000.00	Certified Gynecology Nurse	110	AVLE
122101	Emily	Johnson	F	American	20000.00	Certified Registered Nurse Anesthetist	1208	AVLE
122202	Michael	Garcia	M	American	22000.00	Certified Registered Nurse Anesthetist	1208	AVLE
123701	Ahad	Yousef	F	Saudi	24000.00	Certified Dermatology Nurse	2137	AVLE
123702	Maria	Mazin	F	Saudi	21000.00	Certified Dermatology Nurse	2137	AVLE
124201	Mushal	Musren	M	Saudi	21000.00	Certified Otolaryngology Nurse	2042	AVLE
124202	Sarah	Saleh	F	Saudi	21000.00	Certified Otolaryngology Nurse	2042	AVLE
124401	Fewa	Dracham	F	Saudi	22000.00	Certified Neurology Nurse	6044	AVLE
124402	Selena	Smith	F	American	20000.00	Certified Neurology Nurse	6044	AVLE
124501	Ghadi	Ahmed	F	Saudi	20000.00	Certified Oncology Nurse	3045	AVLE
124502	Alifneel	iyad	F	Saudi	21000.00	Certified Oncology Nurse	3045	AVLE
124601	Mazin	Hmed	M	Saudi	21000.00	Certified Ophthalmic Nurse	46	AVLE
124602	Aran	Abdelaziz	F	Saudi	22000.00	Certified Ophthalmic Nurse	46	AVLE
124801	Drax	Talal	F	Saudi	22000.00	Certified Gastroenterology Nurse	1048	AVLE
124802	Abdelkader	Almen	M	Saudi	25000.00	Certified Gastroenterology Nurse	1048	AVLE
125501	Levi	Abdelkader	M	Saudi	25000.00	Certified Hematology Nurse	2055	AVLE
125502	Muham	Pars	M	Saudi	26000.00	Certified Hematology Nurse	2055	AVLE
125701	Anai	Ameer	F	Saudi	14000.00	Certified Urology Nurse	4057	AVLE
125702	Hazin	Adeeb	M	Saudi	22000.00	Certified Urology Nurse	4057	AVLE
125801	Sophia	Jad	F	Egyptian	22000.00	Certified Cardiac Nurse	58	AVLE
125802	Neda	Osamah	F	Saudi	21000.00	Certified Cardiac Nurse	58	AVLE

36

What is the most needed blood type?

```
SELECT TOP 1 BloodType, COUNT(*) AS NumPatients
FROM Patient
GROUP BY BloodType
ORDER BY COUNT(*) DESC;
```

Results	Messages	
	BloodType	NumPatients
1	0+	11

37

How many families with insurance are registered at the hospital (assuming insurance number uniquely identifies a family)?

```
SELECT InsuranceNumber, COUNT(*) AS FamilyMembers
FROM Patient
WHERE InsuranceNumber NOT LIKE 'No Insurance'
GROUP BY InsuranceNumber
HAVING COUNT(*) > 1;
```

ts Messages

InsuranceNumber	FamilyMembers
111-021-001	2
119-083-321	2
224-443-343	2
083-993-001	2

38

What is the count of doctors for each certification, grouped by certification?

```
SELECT CERTIFICATION, COUNT(*) AS DoctorCount
FROM DOCTOR
GROUP BY CERTIFICATION
ORDER BY CERTIFICATION;
```

Results Messages

	CERTIFICATION	DoctorCount
1	Bachelors degree in Psychology,MSP	1
2	BHSc, Mdent	1
3	MBBS, MD, FRANZCO, FRACS	1
4	Bachelors degree in Psychology,MAP	1
5	BDSc, MDSc, MRACDS (Ortho), FRACDS	1
6	BDSc, MRACDS (GDP)	1
7	BS,MD,DS,FRACS, FAAP	1
8	CNS	1
9	CNS,RDN	1
10	DDS,MSc,PhD	1
11	DO	1
12	DPT	2
13	ENT,MD	2
14	MBBS, FRANZCO	2
15	MD	14
16	MD,ABA	1
17	MD,ABD	1
18	MD,ABIM	2
19	MD,ABU	1
20	MD,BLS	1
21	MD,FAAP	1
22	MD,FRACS	1
23	MD,GYN,REI	1
24	MD,MIGS	1

39

How many patients have allergies to specific products such as milk or eggs?

```
SELECT PatientID , FirstName , LastName , Allergies
FROM Patient WHERE Allergies LIKE '%Milk%' OR Allergies LIKE '%Eggs%';
```

Results Messages

	PatientID	FirstName	LastName	Allergies
1	110106	Dana	Mohamed	Eggs
2	110117	Yousef	Zaid	No Milk
3	110118	Manar	Rayan	Milk
4	110127	Osamah	Husham	Eggs

40

What are the first names, last names, certifications, and department names of nurses?

```
SELECT Nurse.FirstName, Nurse.LastName, Nurse.Certification, Department.DepartmentName
FROM Nurse
FULL JOIN Department ON Nurse.Dept_ID = Department.Dept_ID;
```

FirstName	LastName	Certification	DepartmentName
Sophia	Khan	Certified Psychiatric Nurse	Psychiatry
Arlene	Bachin	Certified Psychiatric Nurse	Psychiatry
Ewa	Gomez	Certified Nutrition and Dietetics Nurse	Nutrition and Dietetics
Emily	Johnson	Certified Nutrition and Dietetics Nurse	Nutrition and Dietetics
Maria	Chang	Certified Dental Nurse	Dentistry
Michael	Nguyen	Certified Dental Nurse	Dentistry
Hadia	Farazi	Certified Physiotherapy Nurse	Physiotherapy
Husham	Hussein	Certified Physiotherapy Nurse	Physiotherapy
Rabe	Ali	Certified Gynecology Nurse	Gynecology
Jawry	Reid	Certified Gynecology Nurse	Gynecology
Emily	Johnson	Certified Registered Nurse Anesthetist	Anesthesiology
Michael	Garcia	Certified Registered Nurse Anesthetist	Anesthesiology
Abad	Youssef	Certified Dermatology Nurse	Dermatology
Maria	Ram	Certified Dermatology Nurse	Dermatology
Mehel	Hussein	Certified Otolaryngology Nurse	Otolaryngology
Sarah	Karim	Certified Otolaryngology Nurse	Otolaryngology
Freda	Shawlan	Certified Neurology Nurse	Neurology
Selma	Bachin	Certified Neurology Nurse	Neurology
Shadi	Alam	Certified Oncology Nurse	Oncology
Alkhair	Eyed	Certified Oncology Nurse	Oncology
Reem	Hussein	Certified Ophthalmology Nurse	Ophthalmology
Aram	Abulhasan	Certified Ophthalmology Nurse	Ophthalmology
Dray	Zaid	Certified Gastroenterology Nurse	Gastroenterology
HadiKhadir	Aliem	Certified Gastroenterology Nurse	Gastroenterology
Levi	Abulbader	Certified Hematology Nurse	Hematology
Husham	Fary	Certified Hematology Nurse	Hematology
Awel	Alwan	Certified Urology Nurse	Urology
Hadi	Abdel	Certified Urology Nurse	Urology
Sophia	Jad	Certified Cardiac Nurse	Cardiology
Nadia	Osamah	Certified Cardiac Nurse	Cardiology
NOEL	NOEL	NOEL	Pharmacy
NOEL	NOEL	NOEL	Admissions
NOEL	NOEL	NOEL	General Services
NOEL	NOEL	NOEL	Finance Department
NOEL	NOEL	NOEL	Human Resources

41

How many patients are there in each department, and what are their respective department names?

```
SELECT d.DepartmentName, COUNT(a.AppointmentDate) AS NumAppointments
FROM Department d
LEFT JOIN Appointments a ON d.Dept_ID = a.Dept_ID
GROUP BY d.DepartmentName
HAVING COUNT(a.AppointmentDate) > 0
ORDER BY NumAppointments DESC;
```

DepartmentName	NumAppointments
Dermatology	7
Dentistry	6
Gastroenterology	5
Physiotherapy	5
Psychiatry	3
Neurology	3
Ophthalmology	3
Cardiology	2
Gynecology	1
Nutrition and Dietetics	1

42

Which male patients have appointments with an oncologist?

```
SELECT DISTINCT P.FirstName, P.LastName
FROM Patient P, Appointments A, Department Dept
WHERE Dept.DepartmentName = 'Oncology' AND A.PatientID = P.PatientID AND P.PatientID NOT IN
( SELECT PatientID
FROM Patient
WHERE Gender = 'F' );
```

FirstName	LastName
Khalid	Abdulahman
Fahad	Abdulaziz
Saleh	Ahmed
Yahya	Bader
Sayed	Fares
Firas	Faisal
Talal	Hamad
Omar	Husam
Hussam	Ibrahim
Youssef	Majed
Haroon	Mala
Saad	Mala
Omar	Samah
Nasser	Salah
Sultan	Salman

43

**Which patients have a Dental
checkup?**

```
--Which patients have a Dental checkup?
SELECT FirstName, LastName
FROM Patient
WHERE PatientID IN (
    SELECT PatientID
    FROM MedicalRecord
    WHERE Diagnosis = 'Dental checkup');
```

ts	Messages
FirstName	LastName
Rand	Fasial
Leena	Ahemad
Fahad	Abdulaziz
Hussam	Ibrahim

44

**What are the details of nurses
who work in the Gynecology
department?**

```
SELECT n.NurseID, n.FirstName, n.LastName, n.Certification
FROM Nurse n
INNER JOIN Department d ON n.Dept_ID = d.Dept_ID
WHERE d.DepartmentName = 'Gynecology';
```

Messages			
NurseID	FirstName	LastName	Certification
123801	Rama	Ali	Certified Gynecology Nurse
123802	Joury	Hani	Certified Gynecology Nurse

45

**Change the salary to
all nurses who work
in Psychiatry to
25000?**

```
UPDATE Nurse
SET salary = 25000
WHERE Dept_ID = 01202;

select *from nurse WHERE Dept_ID = 01202;
```

Its Messages

NurseID	FirstName	LastName	Gender	Nationality	salary	Certification	Dept_ID
120201	Sophia	Khan	F	Pakistani	25000.00	Certified Psychiatric Nurse	1202
120202	Ariana	Smith	F	American	25000.00	Certified Psychiatric Nurse	1202

46

**How many doctors are there in each
department, and what are their
respective department names?**

```
SELECT d.DepartmentName, COUNT(*) AS NumberOfDoctors
FROM Department d
LEFT JOIN Doctor doc ON d.Dept_ID = doc.Dept_ID
GROUP BY d.DepartmentName
HAVING COUNT(*) > 1
ORDER BY NumberOfDoctors DESC;
```

Results Messages

	DepartmentName	NumberOfDoctors
1	Cardiology	5
2	Dentistry	4
3	Dermatology	4
4	Neurology	4
5	Urology	3
6	Ophthalmology	3
7	Otolaryngology	2
8	Physiotherapy	2
9	Psychiatry	2
10	Gastroenterology	2
11	Gynecology	2
12	Hematology	2
13	Anesthetics	2
14	Nutrition and Di	2
15	Oncology	2

47

Set all the salary staff members who work in Admissions department to 15000?

```
----- Set the salary of all staff members who work in Admissions department
```

```
UPDATE Staff
```

```
SET salary = 15000
```

```
WHERE Dept_ID = 00873;
```

```
select * from staff WHERE Dept_ID = 00873;
```

Messages

StaffID	FirstName	LastName	Gender	Nationality	salary	Role	Dept_ID
311901	Nora	Abdullah	F	Saudi	15000.00	Social Work	8873
311902	Zain	Ibrahim	M	Saudi	15000.00	Social Work	8873
311903	Bader	Rakan	M	Jordanian	15000.00	Social Work	8873
311904	Braa	Jad	M	Jordanian	15000.00	Social Work	8873
311905	Eyad	Fars	M	Saudi	15000.00	Receptionist	8873
311906	Aryam	Abdulkareem	F	Saudi	15000.00	Receptionist	8873
311907	Zayd	Mohammed	M	Saudi	15000.00	Receptionist	8873
311908	Kadi	Saud	F	Saudi	15000.00	Receptionist	8873
311909	Rana	Majed	F	Saudi	15000.00	Receptionist	8873
311910	Rahaf	Salem	F	Saudi	15000.00	Receptionist	8873
311911	Raghad	Yousef	F	Saudi	15000.00	Receptionist	8873
311912	Majed	Saleh	M	Saudi	15000.00	Receptionist	8873
311913	Isa	Ahmad	M	Egyptian	15000.00	Receptionist	8873
311914	Malik	Abdullah	M	Saudi	15000.00	Receptionist	8873
311915	Alanoud	Abdulaziz	F	Saudi	15000.00	Receptionist	8873
311916	Musa	Firas	M	Saudi	15000.00	Receptionist	8873
311917	Abdulrahman	Saad	M	Saudi	15000.00	Receptionist	8873
311918	Refal	Tamim	F	Saudi	15000.00	Receptionist	8873
311919	Nawaf	Fasial	M	Saudi	15000.00	Receptionist	8873
311920	Turki	Fahad	M	Saudi	15000.00	administrative assistant	8873
311921	Thamer	Khalid	M	Saudi	15000.00	administrative assistant	8873

48

Who are the male receptionists with salaries between \$8000 and \$9000?

```
select FirstName + ' ' + LastName AS Name, salary
from Staff
where Gender = 'M' and salary between 8000 AND 9000 and role = 'Receptionist';
```

Results Messages

Name	salary
Zayd Mohammed	8100.00
Majed Saleh	8300.00
Musa Firas	8100.00

49

Which time slots have the most appointments in the hospital?

```
SELECT TOP 1 AppointmentTime, COUNT(*) AS AppointmentCount
FROM Appointments
GROUP BY AppointmentTime
ORDER BY AppointmentCount DESC;
```

Results Messages

	AppointmentTime	AppointmentCount
1	11:30:00	4

50

Delete the patient feedback to all the patients who don't have any suggestions

```
919 DELETE FROM PatientFeedback
920 WHERE Suggestions LIKE 'No Suggestions';
```

```
923 select *
924 FROM PatientFeedback
925
```

Results Messages

	FeedbackID	PatientID	SatisfactionRate	Suggestions	FeedbackDate
1	7703	110107	3	Waiting time was a bit long, but overall satisfied w...	1445-08-27
2	7705	110118	2	Would appreciate more detailed explanations from the...	1445-08-19
3	7707	110113	4	Improved signage needed within the hospital.	1445-07-15
4	7708	110115	3	Would like more comfortable waiting areas.	1445-09-20
5	7710	110120	2	Suggest providing more amenities for chemotherapy pa...	1445-07-05

Normalization

Patient

PatientID	FirstNam	LastName	BirthDate	age	Gender	Salutation	PhoneNumber	EmergencyContactName	Allergies	InsuranceNumber	BloodType
110101	Lana	Ahmead	1996-03-03	28	F	Ms	505587433, 509832476	Ahmead Nasser	Peanut	111-021-001	O+
110102	Lamar	Fahad	2006-03-31	18	F	Ms	504599344, 505050000	Waleed Fahad	No Allergies	112-022-221	O+
110103	Naif	Saleh	2012-01-04	12	M	Mr	505170444, 501237895	Saleh Ali	Soya	111-221,920	A+
110104	Leen	Sultan	2006-10-26	17	F	Ms	405735728, 500989893	Kadi Ali	Fish	No Insurance	O+
110105	Sultan	Mohammed	1979-04-26	45	M	Mr	405735728, 502333880	Asma Khalid	No Allergies	No Insurance	B+
110106	Dana	Mohamed	2010-11-09	13	F	Ms	505464952, 500938274	Nehad Abdullah	Eggs	112-989-030	AB+
110107	Hussam	Ali	1989-09-06	34	M	Mr	500630222, 503128474	Btocl Ali	Fish	115-595-001	A+
110108	Arwa	Hamed	2002-02-28	22	F	Ms	506146434, 505545478	Nora Attam	No Allergies	No Insurance	B-
110109	Rand	Fasial	2019-01-30	5	F	Ms	506646433, 5089846352	Fasial Saud	Peanut	221-221-921	O+
110110	Bayan	Salman	1991-08-10	32	F	Ms	505446443, 508376589	Nawaf Fasial	No Allergies	224-443-343	O+
110111	Firas	Fasial	2024-03-28	0	M	Mr	505446443, 505749372	Bayan Salman	No Allergies	224-443-343	O+
110112	Abdullah	Rsheed	1998-12-02	25	M	Mr	568850625, 509812465	Rsheed Talal	Melon	No Insurance	B+
110113	Saleh	Ahemad	1959-01-17	65	M	Mr	504894258, 500955643	Noor Saleh	No Allergies	992-232-040	A+
110114	Luluah	Ibrahim	1938-10-19	85	F	Ms	539404445, 508776542	Bedoor Saleh	No Allergies	119-083-321	AB+
110115	Nasser	Saleh	1980-03-06	44	M	Mr	503023203, 505051735	Nada Fahad	Shellfish	119-083-321	AB+
110116	Leena	Ahemad	2015-07-07	8	F	Ms	538815577, 508799347	Ahemad Khalid	No Allergies	No Insurance	B+
110117	Yousef	Zaid	1972-03-31	52	M	Mr	50180463, 504482614	Maha Naif	Milk	662-910-123	O+
110118	Manar	Rayan	2003-06-04	20	F	Ms	556794717, 507894563	Rayan Abdulaziz	Milk	No Insurance	A-
110119	Omar	Osamah	2008-11-17	15	M	Mr	537665745, 505781298	Osamah Turki	No Allergies	506-911-128	B+
110120	Marah	Saad	2007-02-27	17	F	Ms	505834258, 5057537777	Mona Jamal	Fish	No Insurance	A+
110121	Haya	Rasheed	1978-05-09	45	F	Ms	5074329683, 505822100	Talal Turki	No Allergies	No Insurance	A+
110122	Nawaf	Rakan	2012-01-04	12	M	Mr	501693371, 560472456	Rakan Naif	Fish	222-934-848	B+
110123	Luna	Sultan	2018-07-01	5	F	Ms	505587433, 595826482	Lana Ahemed	No Allergies	111-021-001	B+
110124	Yahya	Bader	2004-04-19	20	M	Mr	505624354, 558550000	Bader Zaid	Sesame Seeds	757-983-003	AB+
110125	Sattam	Saiman	1990-04-22	34	M	Mr	505395532, 583274688	Muneraah Salman	Sesame Seeds	No Insurance	A-
110126	Huda	Tamim	2005-09-22	18	F	Ms	508974693, 555850382	Tamim Nasser	Benanas	No Insurance	B+
110127	Osamah	Husham	1988-12-10	35	M	Mr	505983665, 555685685	Drta Husham	Eggs	No Insurance	A+
110129	Asmaa	Omar	1974-07-19	49	F	Ms	553893275, 505768943	Kadi Ibrahim	No Allergies	983-993-001	AB+
110130	Hussam	Ibrahim	2014-03-05	10	M	Mr	553893275, 505867465	Asmaa Omar	Kiwis	983-993-001	B+
110132	Maha	Ibrahim	1998-11-18	25	F	Ms	557632211, 501377684	Ibrahim Nawaf	Peanut	No Insurance	O+
110133	Khalid	Abdelrahman	1993-07-20	29	M	Mr	505067433, 502212248	Abdelrahman Saud	Peanut	999-878-444	O+
110134	Saud	Fars	1949-01-18	75	M	Mr	505058988, 505767679	Farah Fars	No Allergies	No Insurance	B+
110135	Farah	Isa	2005-12-31	18	F	Ms	505999887, 506622132	Isa Mohammed	Peanut	999-000-123	O+
110136	Marwan	Malk	1992-03-12	32	M	Mr	509871234, 505089443	Malk Mohammed	No Allergies	No Insurance	B+
110137	Lama	Abdullah	2006-10-24	17	F	Ms	508932278, 501238809	Abdullah Saud	Strawberries	876-999-003	AB+
110138	Hussa	Tariq	1970-08-14	53	F	Ms	508974666, 504543812	Dania Fars	No Allergies	No Insurance	O+
110139	Suleiman	Hamed	1997-04-12	27	M	Mr	500057489, 509112004	Hamed Rakan	Strawberries	No Insurance	B-
110140	Saad	Malk	2007-10-23	16	M	Mr	509876432, 504120120	Malk Khalid	Peanut	No Insurance	A+

1NF

The table has a multiple values (has 2 phone numbers)

Patient_1

PatientID	FirstNam	LastName	BirthDate	age	Gender	Salutation	PhoneNumber	PhoneNumber2	EmergencyContactName	Allergies	InsuranceNumber	Blood
110101	Lana	Ahmead	1996-03-03	28	F	Ms	505587433	509832476	Ahmead Nasser	Peanut	111-021-001	O+
110102	Lamar	Fahad	2006-03-31	18	F	Ms	504599344	505050000	Waleed Fahad	No Allergies	112-022-221	O+
110103	Naif	Saleh	2012-01-04	12	M	Mr	505170444	501237895	Saleh Ali	Soya	111-221,920	A+
110104	Leen	Sultan	2006-10-26	17	F	Ms	405735728	500989893	Kadi Ali	Fish	No Insurance	O+
110105	Sultan	Mohammed	1979-04-26	45	M	Mr	405735728	502333880	Asma Khalid	No Allergies	No Insurance	B+
110106	Dana	Mohamed	2010-11-09	13	F	Ms	505464952	500938274	Nehad Abdullah	Eggs	112-989-030	AB+
110107	Hussam	Ali	1989-09-06	34	M	Mr	500630222	503128474	Btool Ali	Fish	115-595-001	A+
110108	Arwa	Hamed	2002-02-28	22	F	Ms	506146434	505545478	Nora Azzam	No Allergies	No Insurance	B-
110109	Rand	Fasial	2019-01-30	5	F	Ms	506646433	5089846352	Fasial Saud	Peanut	221-221-921	O+
110110	Bayan	Salman	1991-08-10	32	F	Ms	505446443	508376589	Nawaf Fasial	No Allergies	224-443-343	O+
110111	Firas	Fasial	2024-03-28	0	M	Mr	505446443	505749372	Bayan Salman	No Allergies	224-443-343	O+
110112	Abdullah	Rsheed	1998-12-02	25	M	Mr	568850625	509812465	Rsheed Talal	Melon	No Insurance	B+
110113	Saleh	Ahemad	1959-01-17	65	M	Mr	504894258	500955643	Noor Saleh	No Allergies	992-232-040	A+
110114	Luluah	Ibrahim	1938-10-19	85	F	Ms	539440445	508776542	Bedoor Saleh	No Allergies	119-083-321	AB+
110115	Nasser	Saleh	1980-03-06	44	M	Mr	503023203	505051735	Nada Fahad	Shellfish	119-083-321	AB+
110116	Leena	Ahemad	2015-07-07	8	F	Ms	538815577	508793547	Ahemad Khalid	No Allergies	No Insurance	B+
110117	Yousef	Zaid	1972-03-31	52	M	Mr	501804638	504482614	Maha Naif	Milk	662-910-123	O+
110118	Manar	Rayan	2003-06-04	20	F	Ms	556794717	507894563	Rayan Abdulaziz	Milk	No Insurance	A-
110119	Omar	Osamah	2008-11-17	15	M	Mr	537665745	505781298	Osamah Turki	No Allergies	506-911-128	B+
110120	Marah	Saad	2007-02-27	17	F	Ms	505834258	505753777	Mona Jamal	Fish	No Insurance	A+
110121	Haya	Rasheed	1978-05-09	45	F	Ms	50743296834	505822100	Talal Turki	No Allergies	No Insurance	A+
110122	Nawaf	Rakan	2012-01-04	12	M	Mr	501693371	560472456	Rakan Naif	Fish	222-934-848	B+
110123	Luna	Sultan	2018-07-01	5	F	Ms	505587433	595826482	Lana Ahemed	No Allergies	111-021-001	B+
110124	Yahya	Bader	2004-04-19	20	M	Mr	505,624,354	558550000	Bader Zaid	Sesame Seeds	757-983-003	AB+
110125	Sattam	Salman	1990-04-22	34	M	Mr	505395532	583274688	Muneraah Salman	Sesame Seeds	No Insurance	A-
110126	Huda	Tamim	2005-09-22	18	F	Ms	508974693	555850382	Tamim Nasser	Bananas	No Insurance	B+
110127	Osamah	Husham	1988-12-10	35	M	Mr	505983665	555685685	Drra Husham	Eggs	No Insurance	A+
110129	Asmaa	Omar	1974-07-19	49	F	Ms	553893275	505768943	Kadi Ibrahim	No Allergies	983-993-001	AB+
110130	Hussam	Ibrahim	2014-03-05	10	M	Mr	553893275	505867465	Asmaa Omar	Kiwi	983-993-001	B+
110132	Maha	Ibrahim	1998-11-18	25	F	Ms	557632211	501377684	Ibrahim Nawaf	Peanut	No Insurance	O+
110133	Khalid	Abdelrahman	1993-07-20	29	M	Mr	505067433	502212248	Abdelrahman Saud	Peanut	999-878-444	O+
110134	Saud	Fars	1949-01-18	75	M	Mr	505058988	505767679	Farah Fars	No Allergies	No Insurance	B+
110135	Farah	Isa	2005-12-31	18	F	Ms	505998887	506622132	Isa Mohammed	Peanut	999-000-123	O+
110136	Marwan	Malk	1992-03-12	32	M	Mr	509871234	505089443	Malk Mohammed	No Allergies	No Insurance	B+
110137	Lama	Abdullah	2006-10-24	17	F	Ms	508932278	501238809	Abdullah Saud	Strawberries	876-999-003	AB+
110138	Hussa	Tariq	1970-08-14	53	F	Ms	508974666	504543812	Dania Fars	No Allergies	No Insurance	O+
110139	Suleiman	Hamed	1997-04-12	27	M	Mr	500057489	509112004	Hamed Rakan	Strawberries	No Insurance	B-
110140	Saad	Malk	2007-10-23	16	M	Mr	509876432	504120120	Malk Khalid	Peanut	No Insurance	A+

2NF

The table has a partial dependency

AgeTable

PatientID	BirthDate	age
110101	1996-03-03	28
110102	2006-03-31	18
110103	2012-01-04	12
110104	2006-10-26	17
110105	1979-04-26	45
110106	2010-11-09	13
110107	1989-09-06	34
110108	2002-02-28	22
110109	2019-01-30	5
110110	1991-08-10	32
110111	2024-03-28	0
110112	1998-12-02	25
110113	1959-01-17	65
110114	1938-10-19	85
110115	1980-03-06	44
110116	2015-07-07	8
110117	1972-03-31	52
110118	2003-06-04	20
110119	2008-11-17	15
110120	2007-02-27	17
110121	1978-05-09	45
110122	2012-01-04	12
110123	2018-07-01	5
110124	2004-04-19	20
110125	1990-04-22	34
110126	2005-09-22	18
110127	1988-12-10	35
110129	1974-07-19	49
110130	2014-03-05	10
110132	1998-11-18	25
110133	1993-07-20	29
110134	1949-01-18	75
110135	2005-12-31	18
110136	1992-03-12	32
110137	2006-10-24	17
110138	1970-08-14	53
110139	1997-04-12	27
110140	2007-10-23	16

Patient_2

PatientID	FirstName	LastName	Gender	Salutation	PhoneNumber	PhoneNumber2	EmergencyContactName	Allergies	InsuranceNumber	BloodType
110101	Lara	Ahmad	F	Ms	505587483	509822476	Ahmed Nasser	Peanut	111-021-801	O+
110102	Lamar	Fahad	F	Ms	504506344	050500000	Waleed Fahad	No Allergies	112-027-121	O+
110103	Nail	Saleh	M	Mr	505170444	501237895	Saleh Ali	Soya	113-221-930	A+
110104	Leen	Sultan	F	Ms	405735728	500989893	Kadi Ali	Fish	No Insurance	O+
110105	Sultan	Mohammed	M	Mr	405735728	502333880	Alma Khalid	No Allergies	No Insurance	B+
110106	Dana	Mohammed	F	Ms	505454952	500938254	Wahad Abdullah	Eggs	112-089-830	AB+
110107	Hussam	Ali	M	Mr	506638222	503128474	Itaad Ali	Fish	115-595-801	A+
110108	Anesa	Hamed	F	Ms	506166434	505165478	Nora Azam	No Allergies	No Insurance	B-
110109	Rand	Faisal	F	Ms	506666483	508846152	Facial Saud	Peanut	273-221-421	O+
110110	Bayan	Salman	F	Ms	505444443	508376589	Nawal Faisal	No Allergies	224-443-343	O+
110111	Firas	Faisal	M	Mr	505444443	505749972	Bayan Salman	No Allergies	224-443-343	O+
110112	Abdullah	Ruheed	M	Mr	564850625	509812465	Ruheed Talal	Melon	No Insurance	B+
110113	Saleh	Ahmed	M	Mr	504894258	500955643	Nasir Saleh	No Allergies	992-232-340	A+
110114	Lulwah	Binsham	F	Ms	534444445	508776543	Badour Saleh	No Allergies	116-083-121	AB+
110115	Nasser	Saleh	M	Mr	503021203	505051735	Nada Fahad	Shellfish	115-083-322	AB+
110116	Leena	Ahmed	F	Ms	518815577	508793547	Ahmed Khalid	No Allergies	No Insurance	B+
110117	Yusef	Zaid	M	Mr	501804038	504482014	Maha Naif	Milk	662-919-123	O+
110118	Manar	Rayas	F	Ms	556794717	507894563	Rayan Abdulaziz	Milk	No Insurance	A-
110119	Omar	Osamah	M	Mr	537605745	505781298	Osamah Turki	No Allergies	506-011-128	B+
110120	Marah	Saad	F	Ms	501834258	505753777	Mona Jamal	Fish	No Insurance	A+
110121	Haya	Ruheed	F	Ms	5074356834	505422100	Talal Turki	No Allergies	No Insurance	A+
110122	Nawal	Rakan	M	Mr	501691371	560474256	Rakan Haif	Fish	272-534-648	B+
110123	Lara	Sultan	F	Ms	505587483	395426482	Lama Ahmed	No Allergies	111-021-901	B+
110124	Yahya	Bader	M	Mr	505424354	508550000	Sadr Zaid	Seaweed Seeds	757-983-003	AB+
110125	Sattam	Salman	M	Mr	505391532	583274688	Muataah Saknan	Seaweed Seeds	No Insurance	A-
110126	Huda	Tamim	F	Ms	508974693	555850382	Tamim Nasser	Bananas	No Insurance	B+
110127	Osamah	Husham	M	Mr	505983665	555685085	Dera Husham	Eggs	No Insurance	A+
110129	Azma	Omar	F	Ms	553891275	505768943	Kadi Ibrahim	No Allergies	983-999-001	AB+
110130	Hussein	Ibrahim	M	Mr	503893275	505867465	Azma Omar	Kiwis	983-999-001	B+
110132	Nadia	Ibrahim	F	Ms	557032211	501377064	Brakham Nawal	Peanut	No Insurance	O+
110133	Khalid	Abdelrahman	M	Mr	505067432	502212248	Abdelrahman Saud	Peanut	995-878-444	O+
110134	Saad	Fari	M	Mr	505053988	505767679	Sarah Fari	No Allergies	No Insurance	B+
110135	Farah	Isa	F	Ms	505998887	506622172	Isa Mohammed	Peanut	995-000-123	O+
110136	Marwan	Malik	M	Mr	509871234	505089443	Mali Mohammed	No Allergies	No Insurance	B+
110137	Lama	Abdullah	F	Ms	508952778	501238809	Abdullah Saud	Strawberries	876-999-903	AB+
110138	Husse	Tariq	F	Ms	506974066	504543812	Dania Fari	No Allergies	No Insurance	O+
110139	Saleman	Hamed	M	Mr	506057489	509112064	Hamed Rakan	Strawberries	No Insurance	B-
110140	Saad	Mak	M	Mr	509874432	504120120	Mali Khalid	Peanut	No Insurance	A+

3NF

The table has a transitive functional dependency
(change in gender may change in salutation)

AgeTable

PatientID	BirthDate	age
110101	1996-03-03	28
110102	2006-03-31	18
110103	2012-01-04	12
110104	2006-10-26	17
110105	1979-04-26	45
110106	2010-11-09	13
110107	1989-09-06	34
110108	2002-02-28	22
110109	2019-01-30	5
110110	1991-08-10	32
110111	2024-03-28	0
110112	1998-12-02	25
110113	1959-01-17	65
110114	1938-10-19	85
110115	1980-03-06	44
110116	2015-07-07	8
110117	1972-03-31	52
110118	2003-06-04	20
110119	2008-11-17	15
110120	2007-02-27	17
110121	1978-05-09	45
110122	2012-01-04	12
110123	2018-07-01	5
110124	2004-04-19	20
110125	1990-04-22	34
110126	2005-09-22	18
110127	1988-12-10	35
110129	1974-07-19	49
110130	2014-03-05	10
110132	1998-11-18	25
110133	1993-07-20	29
110134	1949-01-18	75
110135	2005-12-31	18
110136	1992-03-12	32
110137	2006-10-24	17
110138	1970-08-14	53
110139	1997-04-12	27
110140	2007-10-23	16

Patient_3

PatientID	FirstName	LastName	Gender	PhoneNumber	PhoneNumber2	EmergencyContactName	Allergies	InsuranceNumber	Insurance
110101	Lana	Ahmed	F	505587433	508832476	Ahmed Nasser	Peanut	111-021-001	0+
110102	Lamar	Fahad	F	504599944	505050000	Walied Fahad	No Allergies	112-022-221	0+
110103	Nail	Salih	M	505170464	501237895	Salih Ali	Scya	111-221,920	A+
110104	Leen	Sultan	F	405735728	500989899	Kadi Ali	Fish	No Insurance	0+
110105	Sultan	Mohammed	M	405735728	502333880	Asma Khalid	No Allergies	No Insurance	B+
110106	Dara	Mohamed	F	505464952	500988274	Nehad Abdullah	Eggs	112-980-030	A8+
110107	Hussein	Ali	M	508038022	501178474	Riad Ali	Fish	115-595-001	A+
110108	Ariya	Hamed	F	506146434	50545479	Nora Azzam	No Allergies	No Insurance	B-
110109	Rand	Faisal	F	505646433	5089846352	Faisal Saud	Peanut	221-221-921	0+
110110	Bayan	Salman	F	505646443	508376589	Hawaf Faisal	No Allergies	224-443-943	0+
110111	Firas	Faisal	M	505446443	505740372	Bayan Salman	No Allergies	224-443-943	0+
110112	Alxidu Fah	Reheed	M	568506025	505812465	Reheed Tatal	Meat	No Insurance	B+
110113	Salih	Ahmed	M	504054258	500955643	Noor Saleh	No Allergies	952-232-040	A+
110114	Luluah	Ibrahim	F	513440445	508770542	Bedoor Saleh	No Allergies	119-085-521	A8+
110115	Nasser	Salih	M	508023203	505051236	Hadi Fahad	Shellfish	129-082-221	A8+
110116	Leena	Ahmed	F	513815527	508793547	Ahmed Khalid	No Allergies	No Insurance	B+
110117	Yousef	Zaid	M	501804638	504482614	Maha Nail	Milk	962-910-123	0+
110118	Manar	Rayan	F	555294717	507894563	Rayan Abdulaziz	Milk	No Insurance	A-
110119	Omar	Osamah	M	517665765	508781298	Osamah Turki	No Allergies	506-911-128	B+
110120	Marah	Saad	F	505034258	505753777	Mona Jamal	Fish	No Insurance	A+
110121	Maya	Rashad	F	50761296834	508822100	Talal Turki	No Allergies	No Insurance	A+
110122	Nawaf	Talal	M	501053371	504472450	Talal Nail	Fish	221-934-048	B+
110123	Lana	Sulani	F	505587433	505826482	Lana Ahmed	No Allergies	111-021-001	B+
110124	Yahya	Bader	M	5051424254	505850000	Bader Zaid	Sesame Seeds	757-983-003	A8+
110125	Sattam	Salman	M	505595532	583274688	Muniraah Salman	No Insurance	No Insurance	A-
110126	Huda	Tamim	F	508024603	505850282	Tamim Nasser	Bananas	No Insurance	B+
110127	Ouseih	Hussein	M	505983065	505685685	Oraa Hussein	Eggs	No Insurance	A+
110129	Almaaa	Omar	F	553891275	509788948	Kadi Ibrahim	No Allergies	983-999-001	A8+
110130	Hussein	Ibrahim	M	553891275	505850466	Adnan Omar	Kwini	983-999-001	B+
110132	Maha	Ibrahim	F	557612211	501377684	Ibrahim Nawaf	Peanut	No Insurance	0+
110133	Khalid	Abdelrahman	M	505067433	502282248	Abdelrahman Saud	Peanut	999-878-444	0+
110134	Saud	Fars	M	505058988	505767679	Farah Fars	No Allergies	No Insurance	B+
110135	Farah	isa	F	505999887	506622132	isa Mohammed	Peanut	999-000-123	0+
110136	Marwan	Malik	M	508071234	505089443	Malik Mohammed	No Allergies	No Insurance	B+
110137	Lama	Abdullah	F	508912278	501238809	Abdullah Saud	Strawberries	876-999-003	A8+
110138	Hana	Tarig	F	508974666	504588812	Dania Fars	No Allergies	No Insurance	0+
110139	Sulaiman	Hamed	M	500057489	506112004	Hamed Rakan	Strawberries	No Insurance	B-
110140	Saad	Malik	M	508076432	504120120	Malik Khalid	Peanut	No Insurance	A+

GenderSA

Gender	Salutation
F	Ms
M	Mr

Normalization

Staff

StaffID	FirstName	LastName	Gender	Nationality	Residency	Role	salary	Dept_ID	LanguageSpoken
311901	Nora	Abdullah	F	Saudi	Citizen	Social Work	11000	8873	Arabic,English
311902	Zain	Ibrahim	M	Saudi	Citizen	Social Work	11000	8873	Arabic,English
311903	Bader	Rakan	M	Jordanian	Resident	Social Work	11000	8873	Arabic,English
311904	Braa	Jad	M	Jordanian	Resident	Social Work	11000	8873	Arabic,English
311905	Eyad	Fars	M	Saudi	Citizen	Receptionist	8500	8873	Arabic,English
311906	Aryam	Abdulkareem	F	Saudi	Citizen	Receptionist	8500	8873	Arabic,English
311907	Zayd	Mohammed	M	Saudi	Citizen	Receptionist	8500	8873	Arabic,English
311908	Kadi	Saud	F	Saudi	Citizen	Receptionist	8500	8873	Arabic,English
311909	Rana	Majad	F	Saudi	Citizen	Receptionist	8500	8873	Arabic,English
311910	Rahaf	Salem	F	Saudi	Citizen	Receptionist	8500	8873	Arabic,English
311911	Raghad	Yousef	F	Saudi	Citizen	Receptionist	8500	8873	Arabic,English
311912	Majed	Saleh	M	Saudi	Citizen	Receptionist	8500	8873	Arabic,English
311913	Isa	Ahemad	M	Egyptian	Resident	Receptionist	8500	8873	Arabic,English
311914	Malik	Abdullah	M	Saudi	Citizen	Receptionist	8500	8873	Arabic,English
311915	Alanoud	Abdulaziz	F	Saudi	Citizen	Receptionist	8500	8873	Arabic,English
311916	Musa	Firas	M	Saudi	Citizen	Receptionist	8500	8873	Arabic,English
311917	Abdulrahman	Saad	M	Saudi	Citizen	Receptionist	8500	8873	Arabic,English
311918	Refal	Tamim	F	Saudi	Citizen	Receptionist	8500	8873	Arabic,English
311919	Nawaf	Fasial	M	Saudi	Citizen	Receptionist	8500	8873	Arabic,English
311920	Turki	Fahad	M	Saudi	Citizen	administrative assistant	18000	8873	Arabic,English
311921	Thamer	Khalid	M	Saudi	Citizen	administrative assistant	18000	8873	Arabic,English
311922	Aiman	Hussam	M	Saudi	Citizen	Maintenance Technician	12000	18842	Arabic,English
311923	Bander	Eyad	M	Saudi	Citizen	Security Officer	6000	18842	Arabic,English
311924	Abdullah	Yousef	M	Saudi	Citizen	Security Officer	6000	18842	Arabic,English
311925	Bader	Rakan	M	Saudi	Citizen	Security Officer	6000	18842	Arabic,English
311926	Yasmeen	Hani	F	Egyptian	Resident	Patient Services Assistants	9000	18842	Arabic,English
311927	Farah	Mohannad	F	Jordanian	Resident	Patient Services Assistants	9000	18842	Arabic,English
311928	Deem	Bassam	F	Saudi	Citizen	Patient Services Assistants	9000	18842	Arabic,English
311929	Rami	Mohammed	M	Egyptian	Resident	Patient Services Assistants	9000	18842	Arabic,English
311930	Sarah	Johnson	F	American	Resident	HR Manager	23000	38809	Arabic,English
311931	Jaber	Fahad	M	Saudi	Citizen	HR Coordinator	20000	38809	Arabic,English
311932	Fatema	Hatem	F	Saudi	Citizen	Recruitment Specialist	17000	38809	Arabic,English
311933	Saud	Adeel	M	Saudi	Citizen	Development Officer	16000	38809	Arabic,English
311934	Dona	Tamim	F	Saudi	Citizen	Financial Analyst	20000	28848	Arabic,English
311935	Malk	Ibrahim	F	Saudi	Citizen	Accountant	25000	28848	Arabic,English
311934	Dona	Tamim	F	Saudi	Citizen	Budget Analyst	23000	28848	Arabic,English
311937	Wasan	Adeeb	F	Saudi	Citizen	Pharmacist	36000	661	Arabic,English
311938	Remaz	Amar	F	Saudi	Citizen	Pharmacist	36000	661	Arabic,English
311939	Sultan	Hatem	M	Saudi	Citizen	Pharmacist	36000	661	Arabic,English

1NF

The table's values must be atomic

Satff_1

StaffID	FirstName	LastName	Gender	Nationality	Residency	Role	salary	Dept. ID	languageSpoken	SecondlanguageSpoken
311901	Nora	Abdullah	F	Saudi	Citizen	Social Work	11000	8873	Arabic	English
311902	Zain	Ibrahim	M	Saudi	Citizen	Social Work	11000	8873	Arabic	English
311903	Bader	Rakan	M	Jordanian	Resident	Social Work	11000	8873	Arabic	English
311904	Braa	Jad	M	Jordanian	Resident	Social Work	11000	8873	Arabic	English
311905	Eyad	Fars	M	Saudi	Citizen	Receptionist	8500	8873	Arabic	English
311906	Aryam	Abdulkareem	F	Saudi	Citizen	Receptionist	8500	8873	Arabic	English
311907	Zayd	Mohammed	M	Saudi	Citizen	Receptionist	8500	8873	Arabic	English
311908	Kadi	Saud	F	Saudi	Citizen	Receptionist	8500	8873	Arabic	English
311909	Rana	Majad	F	Saudi	Citizen	Receptionist	8500	8873	Arabic	English
311910	Rahaf	Salem	F	Saudi	Citizen	Receptionist	8500	8873	Arabic	English
311911	Raghad	Yousef	F	Saudi	Citizen	Receptionist	8500	8873	Arabic	English
311912	Majed	Saleh	M	Saudi	Citizen	Receptionist	8500	8873	Arabic	English
311913	Isa	Ahemad	M	Egyptian	Resident	Receptionist	8500	8873	Arabic	English
311914	Malik	Abdullah	M	Saudi	Citizen	Receptionist	8500	8873	Arabic	English
311915	Alanoud	Abdulaziz	F	Saudi	Citizen	Receptionist	8500	8873	Arabic	English
311916	Musa	Firas	M	Saudi	Citizen	Receptionist	8500	8873	Arabic	English
311917	Abdulrahman	Saad	M	Saudi	Citizen	Receptionist	8500	8873	Arabic	English
311918	Refal	Tamim	F	Saudi	Citizen	Receptionist	8500	8873	Arabic	English
311919	Nawaf	Fasial	M	Saudi	Citizen	Receptionist	8500	8873	Arabic	English
311920	Turki	Fahad	M	Saudi	Citizen	administrative assistant	18000	8873	Arabic	English
311921	Thamer	Khalid	M	Saudi	Citizen	administrative assistant	18000	8873	Arabic	English
311922	Aiman	Hussam	M	Saudi	Citizen	Maintenance Technician	12000	18842	Arabic	English
311923	Bander	Eyad	M	Saudi	Citizen	Security Officer	6000	18842	Arabic	English
311924	Abdullah	Yousef	M	Saudi	Citizen	Security Officer	6000	18842	Arabic	English
311925	Bader	Rakan	M	Saudi	Citizen	Security Officer	6000	18842	Arabic	English
311926	Yasmeen	Hani	F	Egyptian	Resident	Patient Services Assistants	9000	18842	Arabic	English
311927	Farah	Mohannad	F	Jordanian	Resident	Patient Services Assistants	9000	18842	Arabic	English
311928	Deem	Bassam	F	Saudi	Citizen	Patient Services Assistants	9000	18842	Arabic	English
311929	Rami	Mohammed	M	Egyptian	Resident	Patient Services Assistants	9000	18842	Arabic	English
311930	Sarah	Johnson	F	American	Resident	HR Manager	23000	38809	English	Spanish
311931	Jaber	Fahad	M	Saudi	Citizen	HR Coordinator	20000	38809	Arabic	English
311932	Fatema	Hatem	F	Saudi	Citizen	Recruitment Specialist	17000	38809	Arabic	English
311933	Saud	Adeel	M	Saudi	Citizen	Development Officer	16000	38809	Arabic	English
311934	Dona	Tamim	F	Saudi	Citizen	Financial Analyst	20000	28848	Arabic	English
311935	Malk	Ibrahim	F	Saudi	Citizen	Accountant	25000	28848	Arabic	English
311934	Dona	Tamim	F	Saudi	Citizen	Budget Analyst	23000	28848	Arabic	English
311937	Wasan	Adeeb	F	Saudi	Citizen	Pharmacist	36000	661	Arabic	English
311938	Remaz	Amar	F	Saudi	Citizen	Pharmacist	36000	661	Arabic	English
311939	Sultan	Hatem	M	Saudi	Citizen	Pharmacist	36000	661	Arabic	English

2NF

The table has a partial dependency

RoleSalary

StaffID	Role	salary	Dept_ID
311901	Social Work	11000	8873
311902	Social Work	11000	8873
311903	Social Work	11000	8873
311904	Social Work	11000	8873
311905	Receptionist	8500	8873
311906	Receptionist	8500	8873
311907	Receptionist	8500	8873
311908	Receptionist	8500	8873
311909	Receptionist	8500	8873
311910	Receptionist	8500	8873
311911	Receptionist	8500	8873
311912	Receptionist	8500	8873
311913	Receptionist	8500	8873
311914	Receptionist	8500	8873
311915	Receptionist	8500	8873
311916	Receptionist	8500	8873
311917	Receptionist	8500	8873
311918	Receptionist	8500	8873
311919	Receptionist	8500	8873
311920	administrative assistant	18000	8873
311921	administrative assistant	18000	8873
311922	Maintenance Technician	12000	18842
311923	Security Officer	6000	18842
311924	Security Officer	6000	18842
311925	Security Officer	6000	18842
311926	Patient Services Assistants	9000	18842
311927	Patient Services Assistants	9000	18842
311928	Patient Services Assistants	9000	18842
311929	Patient Services Assistants	9000	18842
311930	HR Manager	23000	38809
311931	HR Coordinator	20000	38809
311932	Recruitment Specialist	17000	38809
311933	Development Officer	16000	38809
311934	Financial Analyst	20000	28848
311935	Accountant	25000	28848
311934	Budget Analyst	23000	28848
311937	Pharmacist	36000	661
311938	Pharmacist	36000	661
311939	Pharmacist	36000	661

Satff_2

StaffID	First Name	Last Name	Gender	Nationality	Residency	languageSpoken	Second languageSpoken
311901	Nora	Abdullah	F	Saudi	Citizen	Arabic	English
311902	Zain	Ibrahim	M	Saudi	Citizen	Arabic	English
311903	Bader	Rakan	M	Jordanian	Resident	Arabic	English
311904	Braa	Jad	M	Jordanian	Resident	Arabic	English
311905	Eyad	Fars	M	Saudi	Citizen	Arabic	English
311906	Aryam	Abdulkareem	F	Saudi	Citizen	Arabic	English
311907	Zayd	Mohammed	M	Saudi	Citizen	Arabic	English
311908	Kadi	Saud	F	Saudi	Citizen	Arabic	English
311909	Rana	Majad	F	Saudi	Citizen	Arabic	English
311910	Rahaf	Salem	F	Saudi	Citizen	Arabic	English
311911	Raghad	Yousef	F	Saudi	Citizen	Arabic	English
311912	Majed	Saleh	M	Saudi	Citizen	Arabic	English
311913	Isa	Ahemad	M	Egyptian	Resident	Arabic	English
311914	Malik	Abdullah	M	Saudi	Citizen	Arabic	English
311915	Alanoud	Abdulaziz	F	Saudi	Citizen	Arabic	English
311916	Musa	Firas	M	Saudi	Citizen	Arabic	English
311917	Abdulrahman	Saad	M	Saudi	Citizen	Arabic	English
311918	Refal	Tamim	F	Saudi	Citizen	Arabic	English
311919	Nawaf	Fasial	M	Saudi	Citizen	Arabic	English
311920	Turki	Fahad	M	Saudi	Citizen	Arabic	English
311921	Thamer	Khalid	M	Saudi	Citizen	Arabic	English
311922	Aiman	Hussam	M	Saudi	Citizen	Arabic	English
311923	Bander	Eyad	M	Saudi	Citizen	Arabic	English
311924	Abdullah	Yousef	M	Saudi	Citizen	Arabic	English
311925	Bader	Rakan	M	Saudi	Citizen	Arabic	English
311926	Yasmeen	Hani	F	Egyptian	Resident	Arabic	English
311927	Farah	Mohannad	F	Jordanin	Resident	Arabic	English
311928	Deem	Bassam	F	Saudi	Citizen	Arabic	English
311929	Rami	Mohammed	M	Egyptian	Resident	Arabic	English
311930	Sarah	Johnson	F	American	Resident	English	Spanish
311931	Jaber	Fahad	M	Saudi	Citizen	Arabic	English
311932	Fatema	Hatem	F	Saudi	Citizen	Arabic	English
311933	Saud	Adeel	M	Saudi	Citizen	Arabic	English
311934	Dona	Tamim	F	Saudi	Citizen	Arabic	English
311935	Malk	Ibrahim	F	Saudi	Citizen	Arabic	English
311934	Dona	Tamim	F	Saudi	Citizen	Arabic	English
311937	Wasan	Adeeb	F	Saudi	Citizen	Arabic	English
311938	Remaz	Amar	F	Saudi	Citizen	Arabic	English
311939	Sultan	Hatem	M	Saudi	Citizen	Arabic	English

3NF

The table has a transitive functional dependency
(change in nationality may change in residency)

RoleSalary

StaffID	Role	salary	Dept_ID
311901	Social Work	11000	8873
311902	Social Work	11000	8873
311903	Social Work	11000	8873
311904	Social Work	11000	8873
311905	Receptionist	8500	8873
311906	Receptionist	8500	8873
311907	Receptionist	8500	8873
311908	Receptionist	8500	8873
311909	Receptionist	8500	8873
311910	Receptionist	8500	8873
311911	Receptionist	8500	8873
311912	Receptionist	8500	8873
311913	Receptionist	8500	8873
311914	Receptionist	8500	8873
311915	Receptionist	8500	8873
311916	Receptionist	8500	8873
311917	Receptionist	8500	8873
311918	Receptionist	8500	8873
311919	Receptionist	8500	8873
311920	administrative assistant	18000	8873
311921	administrative assistant	18000	8873
311922	Maintenance Technician	12000	18842
311923	Security Officer	6000	18842
311924	Security Officer	6000	18842
311925	Security Officer	6000	18842
311926	Patient Services Assistants	9000	18842
311927	Patient Services Assistants	9000	18842
311928	Patient Services Assistants	9000	18842
311929	Patient Services Assistants	9000	18842
311930	HR Manager	23000	38809
311931	HR Coordinator	20000	38809
311932	Recruitment Specialist	17000	38809
311933	Development Officer	16000	38809
311934	Financial Analyst	20000	28848
311935	Accountant	25000	28848
311934	Budget Analyst	23000	28848
311937	Pharmacist	36000	661
311938	Pharmacist	36000	661
311939	Pharmacist	36000	661

Staff 3

StaffID	First Name	Last Name	Gender	Nationality	Language Spoken	Second Language Spoken
311901	Nora	Abdullah	F	Saudi	Arabic	English
311902	Zain	Ibrahim	M	Saudi	Arabic	English
311903	Bader	Rakan	M	Jordanian	Arabic	English
311904	Braa	Jad	M	Jordanian	Arabic	English
311905	Eyad	Fars	M	Saudi	Arabic	English
311906	Aryam	Abdulkareem	F	Saudi	Arabic	English
311907	Zayd	Mohammed	M	Saudi	Arabic	English
311908	Kadi	Saud	F	Saudi	Arabic	English
311909	Rana	Majad	F	Saudi	Arabic	English
311910	Rahaf	Salem	F	Saudi	Arabic	English
311911	Raghad	Yousef	F	Saudi	Arabic	English
311912	Majed	Saleh	M	Saudi	Arabic	English
311913	Isa	Ahemad	M	Egyptian	Arabic	English
311914	Malik	Abdullah	M	Saudi	Arabic	English
311915	Alanoud	Abdullaziz	F	Saudi	Arabic	English
311916	Musa	Firas	M	Saudi	Arabic	English
311917	Abdulrahman	Saad	M	Saudi	Arabic	English
311918	Refal	Tamim	F	Saudi	Arabic	English
311919	Nawaf	Fasial	M	Saudi	Arabic	English
311920	Turki	Fahad	M	Saudi	Arabic	English
311921	Thamer	Khalid	M	Saudi	Arabic	English
311922	Aiman	Hussam	M	Saudi	Arabic	English
311923	Bander	Eyad	M	Saudi	Arabic	English
311924	Abdullah	Yousef	M	Saudi	Arabic	English
311925	Bader	Rakan	M	Saudi	Arabic	English
311926	Yasmeen	Hani	F	Egyptian	Arabic	English
311927	Farah	Mohannad	F	Jordanian	Arabic	English
311928	Deem	Bassam	F	Saudi	Arabic	English
311929	Rami	Mohammed	M	Egyptian	Arabic	English
311930	Sarah	Johnson	F	American	English	Spanish
311931	Jaber	Fahad	M	Saudi	Arabic	English
311932	Fatema	Hatem	F	Saudi	Arabic	English
311933	Saud	Adeel	M	Saudi	Arabic	English
311934	Dona	Tamim	F	Saudi	Arabic	English
311935	Malik	Ibrahim	F	Saudi	Arabic	English
311934	Dona	Tamim	F	Saudi	Arabic	English
311937	Wasan	Adeeb	F	Saudi	Arabic	English
311938	Remaz	Amar	F	Saudi	Arabic	English
311939	Sultan	Hatem	M	Saudi	Arabic	English

NationalityResidency

Nationality	Residency
Saudi	Citizen
Jordanian	Resident
Egyptian	Resident
American	Resident

Normalization

Department

Dept_Location	DepartmentName	creationdate	DeptAge
Building F4	Ophthalmology,Gastroenterology,Otolaryngology,Oncology, and Neurology	10-11-2023, 2-5-2022 ,2-1-2024,12-21-2021,4-3-2023	0,2,0,2,1
Building F5	Cardiology,Hematology,andUrology	11-9-2021,11/9/2021,1/5/2022	2,2,2
Building F11	Physiotherapy,Dentistry	1/5/2022,7/24/2023	2,0
Building F13	Gynecology,Dermatology	5/3/2022,7/24/2023	1,0
Building F20	Nutrition and Dietetics,Psychiatry	10/9/2022,6/6/2022	1,1
Building F66	Pharmacy	2021-11-09	2
Building F32	Anesthetics	2021-11-09	2
Building F88	Admissions,General Services,Finance Department,Human Resources	11/9/2021,11/9/2021,11/9/2021,11/9/2021	2,2,2,2

1NF

The table's values must be atomic

Department_1

Dept_ID	Dept_Location	DepartmentName	creationdate	DeptAge
46	Building F4	Ophthalmology,	2023-10-11	0
58	Building F5	Cardiology	2021-11-09	2
119	Building F11	Physiotherapy	2022-01-05	2
130	Building F13	Gynecology	2022-05-03	1
203	Building F20	Nutrition and Dietetics	2022-10-09	1
661	Building F66	Pharmacy	2021-11-09	2
1048	Building F4	Gastroenterology	2022-02-05	2
1202	Building F20	Psychiatry	2022-06-06	1
2042	Building F4	Otolaryngology	2024-02-01	0
2055	Building F5	Hematology	2021-11-09	2
3045	Building F4	Oncology	2021-12-21	2
3137	Building F13	Dermatology	2023-07-24	0
3208	Building F32	Anesthetics	2021-11-09	2
4057	Building F5	Urology	2022-01-05	2
6044	Building F4	Neurology	2023-04-03	1
8873	Building F88	Admissions	2021-11-09	2
18842	Building F88	General Services	2021-11-09	2
28848	Building F88	Finance Department	2021-11-09	2
38809	Building F88	Human Resources	2021-11-09	2

2NF

The table has a partial dependency

Department_2

Dept_ID	Dept_Location	DepartmentName
46	Building F4	Ophthalmology,
58	Building F5	Cardiology
119	Building F11	Physiotherapy
130	Building F13	Gynecology
203	Building F20	Nutrition and Dietetics
661	Building F66	Pharmacy
1048	Building F4	Gastroenterology
1202	Building F20	Psychiatry
2042	Building F4	Otolaryngology
2055	Building F5	Hematology
3045	Building F4	Oncology
3137	Building F13	Dermatology
3208	Building F32	Anesthetics
4057	Building F5	Urology
6044	Building F4	Neurology
8873	Building F88	Admissions
18842	Building F88	General Services
28848	Building F88	Finance Department
38809	Building F88	Human Resources

departmentDate

Dept_ID	creationdate	DeptAge
46	2023-10-11	0
58	2021-11-09	2
119	2022-01-05	2
130	2022-05-03	1
203	2022-10-09	1
661	2021-11-09	2
1048	2022-02-05	2
1202	2022-06-06	1
2042	2024-02-01	0
2055	2021-11-09	2
3045	2021-12-21	2
3137	2023-07-24	0
3208	2021-11-09	2
4057	2022-01-05	2
6044	2023-04-03	1
8873	2021-11-09	2
18842	2021-11-09	2

3NF

The table has a transitive functional dependency

Department_2

Dept_ID	Dept_Location	DepartmentName
46	Building F4	Ophthalmology,
58	Building F5	Cardiology
119	Building F11	Physiotherapy
130	Building F13	Gynecology
203	Building F20	Nutrition and Dietetics
661	Building F66	Pharmacy
1048	Building F4	Gastroenterology
1202	Building F20	Psychiatry
2042	Building F4	Otolaryngology
2055	Building F5	Hematology
3045	Building F4	Oncology
3137	Building F13	Dermatology
3208	Building F32	Anesthetics
4057	Building F5	Urology
6044	Building F4	Neurology
8873	Building F88	Admissions
18842	Building F88	General Services
28848	Building F88	Finance Department
28809	Building F88	Human Resources

DepartmentDate1

creationdate	DeptAge
2023-10-11	0
2021-11-09	2
2022-01-05	2
2022-05-03	1
2022-10-09	1
2022-02-05	2
2022-06-06	1
2024-02-01	0
2021-12-21	2
2023-07-24	0
2023-04-03	1

DepartmentDate2

Dept_ID	creationdate
46	2023-10-11
58	2021-11-09
119	2022-01-05
130	2022-05-03
203	2022-10-09
661	2021-11-09
1048	2022-02-05
1202	2022-06-06
2042	2024-02-01
2055	2021-11-09
3045	2021-12-21
3137	2023-07-24
3208	2021-11-09
4057	2022-01-05
6044	2023-04-03
8873	2021-11-09
18842	2021-11-09
28848	2021-11-09
28809	2021-11-09